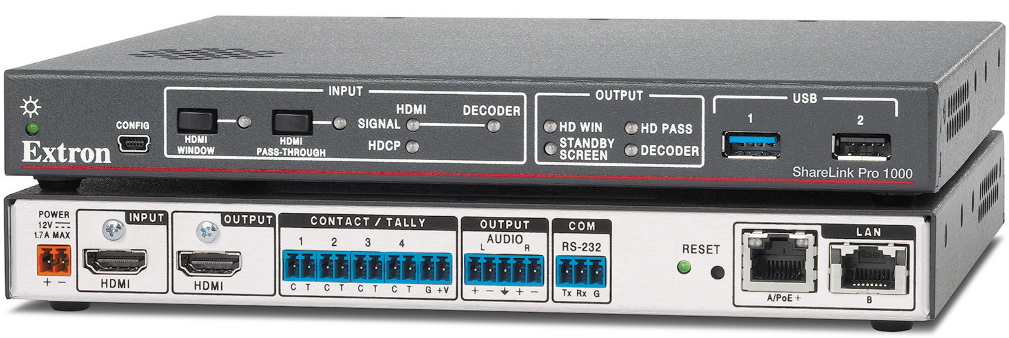
User Guide

Collaboration Systems

ShareLink Pro 1000

Wireless and Wired Collaboration Gateway



Extron

68-3353-01 Rev. D
11 20

Safety Instructions

Safety Instructions • English

⚠ **WARNING:** This symbol, ⚠, when used on the product, is intended to alert the user of the presence of uninsulated dangerous voltage within the product's enclosure that may present a risk of electric shock.

ATTENTION: This symbol, ⚠, when used on the product, is intended to alert the user of important operating and maintenance (servicing) instructions in the literature provided with the equipment.

For information on safety guidelines, regulatory compliances, EMI/EMF compatibility, accessibility, and related topics, see the Extron Safety and Regulatory Compliance Guide, part number 68-290-01, on the Extron website, **www.extron.com**.

Sicherheitsanweisungen • Deutsch

WARNUNG: Dieses Symbol ⚠ auf dem Produkt soll den Benutzer darauf aufmerksam machen, dass im Inneren des Gehäuses dieses Produktes gefährliche Spannungen herrschen, die nicht isoliert sind und die einen elektrischen Schlag verursachen können.

VORSICHT: Dieses Symbol ⚠ auf dem Produkt soll dem Benutzer in der im Lieferumfang enthaltenen Dokumentation besonders wichtige Hinweise zur Bedienung und Wartung (Instandhaltung) geben.

Weitere Informationen über die Sicherheitsrichtlinien, Produkthandhabung, EMI/EMF-Kompatibilität, Zugänglichkeit und verwandte Themen finden Sie in den Extron-Richtlinien für Sicherheit und Handhabung (Artikelnummer 68-290-01) auf der Extron-Website, **www.extron.com**.

Instrucciones de seguridad • Español

ADVERTENCIA: Este símbolo, ⚠, cuando se utiliza en el producto, avisa al usuario de la presencia de voltaje peligroso sin aislar dentro del producto, lo que puede representar un riesgo de descarga eléctrica.

ATENCIÓN: Este símbolo, ⚠, cuando se utiliza en el producto, avisa al usuario de la presencia de importantes instrucciones de uso y mantenimiento recogidas en la documentación proporcionada con el equipo.

Para obtener información sobre directrices de seguridad, cumplimiento de normativas, compatibilidad electromagnética, accesibilidad y temas relacionados, consulte la Guía de cumplimiento de normativas y seguridad de Extron, referencia 68-290-01, en el sitio Web de Extron, **www.extron.com**.

Instructions de sécurité • Français

AVERTISSEMENT : Ce pictogramme, ⚠, lorsqu'il est utilisé sur le produit, signale à l'utilisateur la présence à l'intérieur du boîtier du produit d'une tension électrique dangereuse susceptible de provoquer un choc électrique.

ATTENTION : Ce pictogramme, ⚠, lorsqu'il est utilisé sur le produit, signale à l'utilisateur des instructions d'utilisation ou de maintenance importantes qui se trouvent dans la documentation fournie avec le matériel.

Pour en savoir plus sur les règles de sécurité, la conformité à la réglementation, la compatibilité EMI/EMF, l'accessibilité, et autres sujets connexes, lisez les informations de sécurité et de conformité Extron, réf. 68-290-01, sur le site Extron, **www.extron.com**.

Istruzioni di sicurezza • Italiano

AVVERTENZA: Il simbolo, ⚠, se usato sul prodotto, serve ad avvertire l'utente della presenza di tensione non isolata pericolosa all'interno del contenitore del prodotto che può costituire un rischio di scosse elettriche.

ATTENTZIONE: Il simbolo, ⚠, se usato sul prodotto, serve ad avvertire l'utente della presenza di importanti istruzioni di funzionamento e manutenzione nella documentazione fornita con l'apparecchio.

Per informazioni su parametri di sicurezza, conformità alle normative, compatibilità EMI/EMF, accessibilità e argomenti simili, fare riferimento alla Guida alla conformità normativa e di sicurezza di Extron, cod. articolo 68-290-01, sul sito web di Extron, **www.extron.com**.

Instrukcje bezpieczeństwa • Polska

OSTRZEŻENIE: Ten symbol, ⚠, gdy używany na produkt, ma na celu poinformować użytkownika o obecności izolowanego i niebezpiecznego napięcia wewnątrz obudowy produktu, który może stanowić zagrożenie porażenia prądem elektrycznym.

UWAGI: Ten symbol, ⚠, gdy używany na produkt, jest przeznaczony do ostrzegania użytkownika ważne operacyjne oraz instrukcje konserwacji (obsługi) w literaturze, wyposażone w sprzęt.

Informacji na temat wytycznych w sprawie bezpieczeństwa, regulacji wzajemnej zgodności, zgodność EMI/EMF, dostępności i Tematy pokrewne, zobacz Extron bezpieczeństwa i regulacyjnego zgodności przewodnik, część numer 68-290-01, na stronie internetowej Extron, **www.extron.com**.

Инструкция по технике безопасности • Русский

ПРЕДУПРЕЖДЕНИЕ: Данный символ, ⚠, если указан на продукте, предупреждает пользователя о наличии неизолированного опасного напряжения внутри корпуса продукта, которое может привести к поражению электрическим током.

ВНИМАНИЕ: Данный символ, ⚠, если указан на продукте, предупреждает пользователя о наличии важных инструкций по эксплуатации и обслуживанию в руководстве, прилагаемом к данному оборудованию.

Для получения информации о правилах техники безопасности, соблюдении нормативных требований, электромагнитной совместимости (ЭМП/ЭДС), возможности доступа и других вопросах см. руководство по безопасности и соблюдению нормативных требований Extron на сайте Extron: , **www.extron.com**, номер по каталогу - 68-290-01.

安全说明 • 简体中文

警告: ⚠产品上的这个标志意在警告用户该产品机壳内有暴露的危险 电压，有触电危险。

注 意: ⚠产品上的这个标志意在提示用户设备随附的用户手册中有重要的操作和维护(维修) 说明。

关于我们产品的安全指南、遵循的规范、EMI/EMF 的兼容性、无障碍使用的特性等相关内容，敬请访问 Extron 网站，**www.extron.com**，参见 Extron 安全规范指南，产品编号 68-290-01。

安全記事 • 繁體中文

警告: 若產品上使用此符號，是為了提醒使用者，產品機殼內存在著可能會導致觸電之風險的未絕緣危險電壓。

注意 若產品上使用此符號，是為了提醒使用者，設備隨附的用戶手冊中有重要的操作和維護（維修）說明。

有關安全性指導方針、法規遵守、EMI/EMF 相容性、存取範圍和相關主題的詳細資訊，請瀏覽 Extron 網站：**www.extron.com**，然後參閱《Extron 安全性與法規遵守手冊》，準則編號 68-290-01。

安全上のご注意 • 日本語

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注意: この記号 が製品上に表示されている場合は、本機の取扱説明書に記載されている重要な操作と保守（整備）の指示についてユーザーの注意を喚起するものです。

安全上のご注意、法規厳守、EMI/EMF適合性、その他の関連項目については、エクストロンのウェブサイト **www.extron.com** より『Extron Safety and Regulatory Compliance Guide』(P/N 68-290-01) をご覧ください。

안전 지침 • 한국어

경고: 이 기호 가 제품에 사용될 경우, 제품의 인클로저 내에 있는 접지되지 않은 위험한 전류로 인해 사용자가 감전될 위험이 있음을 경고합니다.

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안전 가이드라인, 규제 준수, EMI/EMF 호환성, 접근성, 그리고 관련 항목에 대한 자세한 내용은 Extron 웹 사이트(**www.extron.com**)의 Extron 안전 및 규제 준수 안내서, 68-290-01 조항을 참조하십시오.

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FCC Class B Notice

NOTE: This device complies with part 15 of the FCC rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) This device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC rules. These limits provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. There is no guarantee that interference will not occur. If this equipment does cause interference to radio or television reception, which can be determined by turning the equipment off and on, you are encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

In order to maintain compliance with FCC regulations, shielded cables must be used with this equipment. Operation with non-approved equipment or unshielded cables is likely to result in interference to radio and TV reception. The user is cautioned that changes and modifications made to the equipment without the approval of the manufacturer could void the user's authority to operate this equipment.

Battery Notice

This product contains a battery. Do not open the unit to replace the battery. If the battery needs replacing, return the entire unit to Extron (for the correct address, see the Extron Warranty section on the last page of this guide).

CAUTION: Risk of explosion. Do not replace the battery with an incorrect type. Dispose of used batteries according to the instructions.

ATTENTION : Risque d'explosion. Ne pas remplacer la pile par le mauvais type de pile. Débarrassez-vous des piles usagées selon le mode d'emploi.

Conventions Used in this Guide

Notifications

The following notifications are used in this guide:

ATTENTION:
- Risk of property damage.
- Risque de dommages matériels.

NOTE: A note draws attention to important information.

Software Commands

Commands are written in the fonts shown here:

```
^AR Merge Scene,,Op1 scene 1,1 ^B 51 ^W^C
[01] R 0004 00300 00400 00800 00600 [02] 35 [17] [03]
```

Esc X1 *X17* X20* X23* X21 CE ←

NOTE: For commands and examples of computer or device responses mentioned in this guide, the character "0" is used for the number zero and "O" is the capital letter "o."

Computer responses and directory paths that do not have variables are written in the font shown here:

```
Reply from 208.132.180.48: bytes=32 times=2ms TTL=32
C:\Program Files\Extron
```

Variables are written in slanted form as shown here:

```
ping xxx.xxx.xxx.xxx —t
SOH R Data STX Command ETB ETX
```

Selectable items, such as menu names, menu options, buttons, tabs, and field names are written in the font shown here:

From the **File** menu, select **New**.

Click the **OK** button.

Specifications Availability

Product specifications are available on the Extron website, **www.extron.com**.

Contents

Introduction

This section covers the following:

- **About the ShareLink Pro 1000**
- **Features**

About the ShareLink Pro 1000

The ShareLink Pro 1000 Wireless and Wired Collaboration Gateway enables anyone to present wireless or wired content from their computers, tablets, or smartphones onto a display for easy collaboration.

It features streaming technology that supports simultaneous display of up to four content sources, including an HDMI-connected device. The HDMI input supports wired connections from any connected source in the room. To support a wide range of environments, the ShareLink Pro 1000 has collaboration and moderator modes that facilitate both open and restrictive environments.

The ShareLink Pro 1000 provides easy integration of AV and mobile devices into meeting, huddle, collaboration, and presentation spaces.

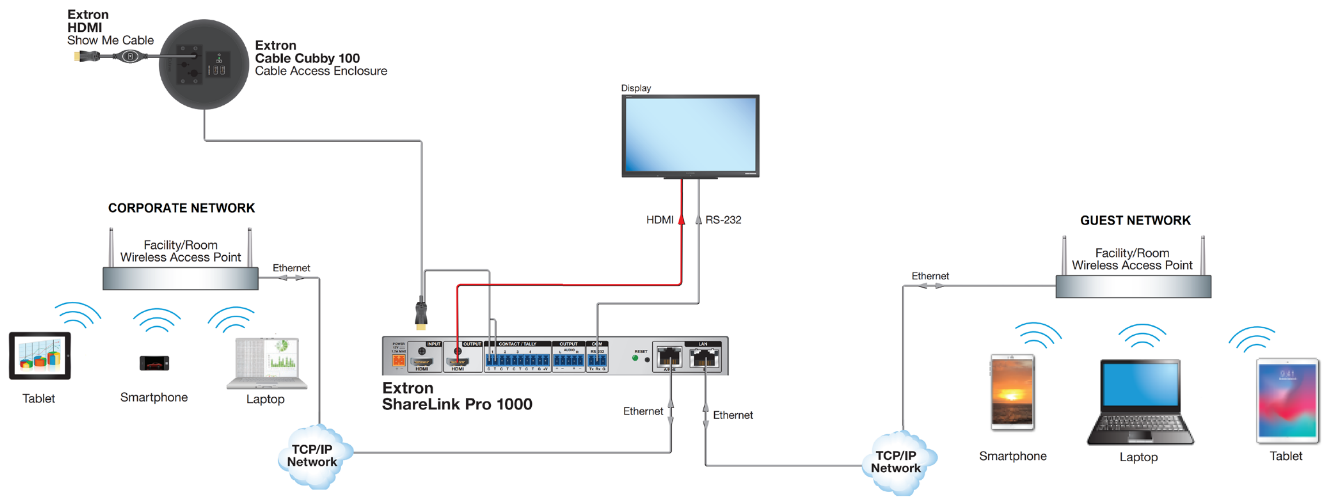


Figure 1. ShareLink Pro 1000 Application Diagram

Features

- **Wirelessly share content from mobile devices** — Connects a wide variety of devices to the system wirelessly or through a wired LAN connection.
- **Provides full screen mirroring for all devices** — Display the entire screen of your device through the wireless collaboration gateway for more fluid and easy collaboration sessions.
- **Wireless and wired sources can collaborate simultaneously** — HDMI input enables wired users or sources to collaborate simultaneously with wireless users in the same session.
- **Supports Mac and Windows computers as well as Apple and Android tablets and smartphones**
- **Dedicated app provides consistent user experience across platforms** — Similar interface for all platforms makes it easy for users to assist one another or move between devices regardless of how they connect and share content.
- **View up to four pieces of content simultaneously from any connected user** — Fosters collaboration and discussion by comparing multiple sources concurrently and reduces or eliminates the need to switch between sources for more effective collaboration.
- **HDMI output supports computer and video resolutions up to 4K** — Resolutions up to 4096x2160 with 4:4:4 chroma subsampling at 8 bits of color.
- **Compatible with TeamWork Show Me cables** — Show Me cables provide convenient connectivity and user input selection and control for TeamWork collaboration systems. Visit the TeamWork System Builder to create a customized system for your collaboration environment.
- **HDCP compliant** — Ensures display of content-protected media on HDMI input and output for interoperability with other HDCP-compliant devices.
- **EDID Minder automatically manages EDID communication between HDMI connected devices** — EDID Minder ensures that all sources power up properly and reliably output content for display.
- **Collaboration mode allows any attendee to display content and control the presentation** — Enables content display from any connected device to enhance interactivity in brainstorming sessions, team meetings, and other collaborative environments.
- **Moderator mode ensures only approved user content is displayed** — Enables the moderator to select which users can access the display and how the content is displayed.
- **Display codes ensure content is delivered only to the selected display devices** — Randomly generated or user-defined display authentication codes prevent unintentional sharing or display of content to an adjacent space.
- **WebView technology displays slide images on attendee's personal devices via a Web browser** — The ShareLink Pro 1000 enables meeting content to display on a participant's mobile device. This is ideal for attendees who cannot easily view the main display.
- **128-bit data encryption** — A variety of security protocols ensure that all content transmitted between devices and the ShareLink Pro 1000 is fully encrypted and secure.
- **Power over Ethernet (PoE+) allows the ShareLink Pro to receive power and communication over a single Ethernet cable, eliminating the need for a local power supply**

- **Dual Gigabit Ethernet** — Provides two high-speed data links, enabling segmentation of guest and private networks for fast and easy access to the web or other network resources.
- **Fully customizable welcome screen** — Multiple configuration options to show, hide, or customize information on the welcome screen, so users can quickly connect and begin sharing their content.
- **Connects to an existing wireless network** — Leverages an existing wireless network infrastructure for the connection of mobile devices.
- **Includes user-friendly software for screen mirroring, sharing, and advanced control** — Provides both an executable that does not require administrator access rights and a version that can deployed by IT administrators.
- **Video screen saver** — The ShareLink Pro 1000 can be set to automatically mute video and sync output to the display device when no active connections are detected, automatically entering the display into standby mode to conserve energy, reducing costs and promoting panel life.
- **Front panel security lockout** — Prevents unauthorized use in non-secure environments. Set via PCS software.
- **Easy setup and commissioning with Extron Product Configuration Software (PCS)** — Conveniently configure multiple products using a single software application.
- **Display control options including RS-232 and CEC over HDMI connection**
- **Contact closure remote control with tally output** — Allows for automatic operation by an optional occupancy sensor or remote selection of the input channel by a button panel. +5 VDC is provided to light an LED to indicate the currently selected input.
- **Automatic input cable equalization to 50 feet (15 meters)** — Actively conditions incoming HDMI signals to compensate for signal loss when using long HDMI cables, low quality HDMI cables, and source devices with poor HDMI signal output.
- **Compatible with Extron mounting solutions** — Easy mounting with Extron low-profile and under-desk mounts, under-table kits, and rack-shelf options.
- **1-inch (2.5 cm) high, half rack width metal enclosure** — Compact, low profile enclosure allows discreet installation within a lectern or behind a flat panel display.
- **Energy-efficient, external universal power supply included** — Provides worldwide power compatibility.
- **Extron Pro Series Secure Control** — The ShareLink Pro 1000 is designed to work natively with Extron Pro Series control systems via its dedicated Secure Platform Interface that provides encrypted communication at all times between an Extron Pro Series control processor and the ShareLink Pro 1000 Collaboration Gateway.

Installation

This section covers the following:

- **Rear Panel Features and Cabling**
- **Front Panel Features and Cabling**
- **Using the Standby Screen**

Rear Panel Features and Cabling

The following section covers rear panel features and cabling procedures. If mounting is necessary before connecting cables, see **Mounting** on page 51.

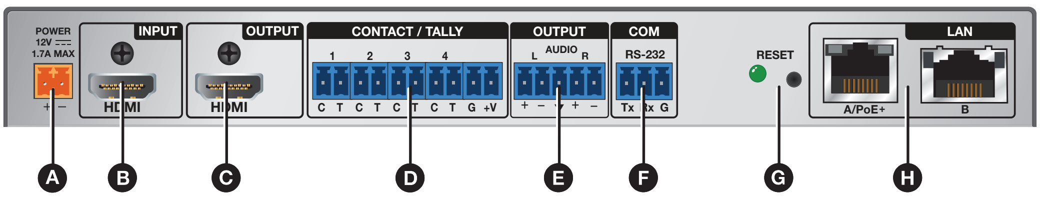


Figure 2. **Rear Panel**

A **Power input** — Connect the provided power supply to this two-pole captive screw connector (see the **Power Notifications** on the next page for important information).

> **NOTES:**
> - Alternatively, the LAN A/PoE+ port can be used to power the ShareLink Pro 1000 via a Power over Ethernet (PoE+) source device.
> - If both power options are connected (12 V power supply and PoE+), PoE+ has precedence.

B **HDMI input** — Connect an HDMI source device to this input.

> **NOTE:** For HDMI connections (B and C), use an Extron LockIt Cable Lacing Bracket to secure the HDMI connector to the device (see **HDMI Connection** page 6).

C **HDMI output** — Connect an HDMI display device to this female HDMI connector.

D **Contact/Tally Ports** — (Optional) Connect push-button contact closure devices, such as Show Me cables, to these connectors to enable input switching via contact closure. Additionally, connect an OCS 100 occupancy sensor to automatically control the display to power on and off.

E **Audio output** — Connect a balanced or unbalanced stereo (or dual mono) line level audio device to this 5-pole 3.5 mm captive screw connector. Wire the connector as shown below:

> **ATTENTION:**
> - For unbalanced audio, connect the sleeves to the ground contact. DO NOT connect the sleeves to the negative (–) contacts.
> - Pour l'audio asymétrique, connectez les manchons au contact au sol. Ne PAS connecter les manchons aux contacts négatifs (–).

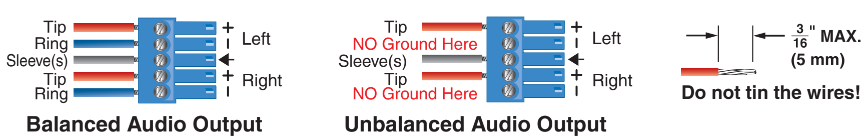


Figure 3. **Audio Output Captive Screw Connector Wiring**

F **COM port (RS-232 endpoint control)** — Connect an RS-232 capable endpoint, such as a display, to this 3-pole captive screw connector.

G **Reset button and LED** — Use this button to reset the device (see **Reset Modes** on page 7 for details on the various resets and LED states).

H **LAN ports** — Use an Ethernet cable to connect a network switch, hub, router, or PC to these female RJ-45 connectors. The LAN A/PoE+ port can also be used to power the ShareLink Pro via a Power over Ethernet (PoE+) source device.

> **NOTE:**
>
> **Default protocol**:
>
> - IP address - LAN A: 192.168.253.254
> - IP address - LAN B: 192.168.254.254
> - Gateway IP address: 0.0.0.0
> - Subnet mask: 255.255.255.0
> - DNS address: 0.0.0.0
> - DHCP: **LAN A**: On, **LAN B**: Off
> - Link speed and duplex level: autodetected
>
> **LED indicators**:
>
> - Left LED (amber) indicates link speed:
>   **1 blink per second**: 10M
>   **2 blinks per second**: 100M
>   **3 blinks per second**: 1G
> - Right LED (green) indicates a link is active

Power Notifications

> **ATTENTION:**
>
> - If not provided with a power supply, this product is intended to be supplied by a UL Listed power source marked “Class 2” or “LPS” and rated output 12Vdc, minimum 2.0A, or 56 Vdc (PoE+), minimum 0.8 A.
> - Si le produit n’est pas fourni avec une source d’alimentation, il doit être alimenté par une source d’alimentation certifiée UL de classe 2 ou LPS, avec une tension nominale 12 Vdc, 2,0 A minimum, ou 56 Vdc (PoE+), 0,8 A minimum.
> - Unless otherwise stated, the AC/DC adapters are not suitable for use in air handling spaces or in wall cavities.
> - Sauf mention contraire, les adaptateurs AC/DC ne sont pas appropriés pour une utilisation dans les espaces d’aération ou dans les cavités murales.
> - The installation must always be in accordance with the applicable provisions of National Electrical Code ANSI/NFPA 70, article 725 and the Canadian Electrical Code part 1, section 16. The power supply shall not be permanently fixed to a building structure or similar structure.
> - Cette installation doit toujours être en accord avec les mesures qui s’applique au National Electrical Code ANSI/NFPA 70, article 725, et au Canadian Electrical Code, partie 1, section 16. La source d’alimentation ne devra pas être fixée de façon permanente à une structure de bâtiment ou à une structure similaire.
> - Power over Ethernet (PoE) is intended for indoor use only. It is to be connected only to networks or circuits that are not routed to the outside plant or building.
> - L’alimentation via Ethernet (PoE) est destinée à une utilisation en intérieur uniquement. Elle doit être connectée seulement à des réseaux ou des circuits qui ne sont pas routés au réseau ou au bâtiment extérieur.

HDMI Connection

To secure the HDMI cable to the HDMI input connector, use an Extron LockIt Cable Lacing Bracket and a tie wrap.

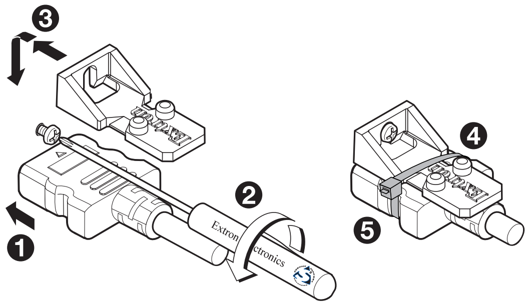


Figure 4. Installing the LockIt Cable Lacing Bracket

1. Plug the HDMI cable into the panel connection (see figure 4, ❶).
2. Loosen the HDMI connection mounting screw from the panel (❷) enough to allow the LockIt to be placed over it. The screw does not have to be removed.
3. Place the LockIt on the screw and against the HDMI connector (❸), and then tighten the screw to secure the bracket.
4. Loosely place the included tie wrap around the HDMI connector and the LockIt (❹).
5. While holding the connector securely against the cable lacing bracket, use pliers or similar tools to tighten the tie wrap, then remove any excess length (❺).

ATTENTION:

- Connect and pull the tie wraps until they are secure. Do not overtighten.
- Connectez et tirez les serre-câbles jusqu'à ce qu'ils soient sécurisés. Ne pas trop serrer.

Reset Modes

The rear panel Reset button can activate four different reset modes (see the table below for a summary of each reset mode and LED state).

NOTE: Review the reset modes carefully. Some reset modes delete all user-loaded content.

Reset Mode Summary

Mode	Activation	Result
Factory Firmware Reset (mode 1) Resets the firmware to the factory version temporarily if an incompatibility issue arises with the current firmware.	Hold down the recessed `Reset` button while applying power to the device. **NOTE:** After this reset, update the device with the latest firmware version. Do not operate the device with the firmware version that results from this reset.	The device reverts to the factory default firmware for a single power cycle. All user files and settings are maintained. **NOTE:** If this reset was made by mistake or is no longer desired, cycle power to the device again to restore the firmware version running prior to the reset.
Reset Factory Defaults (mode 3) Resets all device settings to default values but retains user-loaded files and IP settings. **NOTE:** This reset is the equivalent of SIS command `ZXXX` (see **page 43**)	Hold down the `Reset` button (for about 3 seconds) until the Reset LED blinks once. Then, press the `Reset` button again momentarily (<1 second).	The device reverts all settings to the default values except: • IP related settings • User-loaded files and configurations **NOTE:** • This mode does not affect passwords. • The Reset LED blinks four times in quick succession during the reset, then remains on.
IP Settings Reset (mode 4) Resets the IP settings back to factory defaults.	Hold down the `Reset` button for about 6 seconds until the Reset LED blinks twice (once at 3 seconds, then twice at 6 seconds). Then, within one second, press the `Reset` button again momentarily (for <1 second). **NOTE:** Nothing happens if the button is not pressed within one second.	The device reverts IP settings to the default values. • ARP capability is enabled. • IP address, subnet mask, gateway address, and port mapping reset to the default value. • DHCP is enabled for LAN A, and disabled for LAN B. • Events are disabled. **NOTE:** The Reset LED blinks four times in quick succession during the reset, then remains on.
Reset to Factory Defaults (mode 5) Resets the device to default configuration. **NOTE:** This reset is the equivalent of SIS command `ZQQQ` (see **page 43**)	Hold down the `Reset` button for about 9 seconds until the Reset LED blinks three times (once at 3 seconds, twice at 6 seconds, and thrice at 9 seconds). Then, within one second, press the `Reset` button again momentarily (for <1 second). **NOTE:** Nothing happens if the button is not pressed within one second.	The device reverts user settings to the factory default values (firmware excluded). • All user modifiable configuration is reset to default values, including IP settings and real-time adjustments. • All user loaded files are deleted. **NOTE:** • Admin password resets to `extron` • The Reset LED blinks four times in quick succession during the reset, then remains on.

Front Panel Features and Cabling

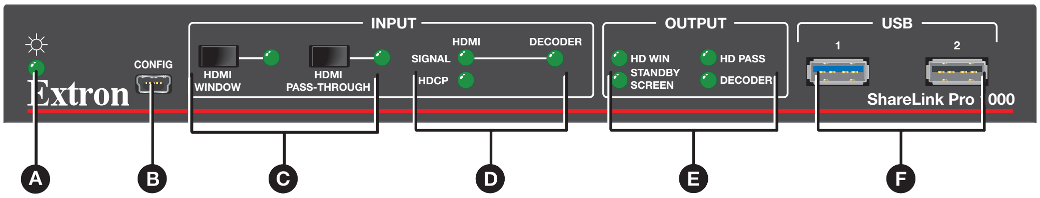


Figure 5. **Front Panel**

Ⓐ **Power LED indicator** — Lights green when the unit is powered on.

Ⓑ **Config USB port** (future use)

Ⓒ **HDMI input selection buttons** — Use these buttons to select between HDMI window mode and pass-through mode. These modes determine how the HDMI input is presented on the output (see below for details).

> **NOTE:** Selection of one mode deselects the other mode.

- `HDMI Window` button – Activates HDMI window mode, which displays the HDMI input source in addition to or alongside any content currently being shared by client devices.

  > **NOTE:** Maximum output resolution for this mode is 4K @ 30Hz.

- `HDMI Pass-Through` button – Activates HDMI pass-through mode, which routes the HDMI input signal directly to the HDMI output unaltered. In this mode, neither the status bar nor decoder input streams are shown on the output.

  > **NOTE:** Maximum output resolution for this mode is 4K @ 60Hz.

Ⓓ **Input LED indicators**:

- **HDMI Signal LED** – Lights green when an HDMI signal is detected.
- **Decoder Signal LED** – Lights green when one stream is detected. Flashes green when more than one stream is detected at the same time.
- **HDCP LED** – Lights green when the source signal is encrypted. Turns off when input signal is not encrypted or when HDCP authorization is disabled.

Ⓔ **Output LED indicators** — These LEDs light green when the source signals are present and active on the output.

- **HD Win** – Lights green when the HDMI source is active on the output in HDMI window mode. Lights amber if HDMI pass-through mode is activated after originally being in window mode.
- **Decoder** – Lights green when at least one stream is active on the output. Lights amber if HDMI pass-through mode is activated after at least one stream was active on the output.
- **Standby Screen** – Lights green when the standby screen is active on the output. Lights amber if HDMI pass-through mode is activated while the ShareLink Pro output displays the standby screen.

  > **NOTE:** The standby screen images are displayed only when HDMI and decoder sources are **not** active.

- **HD Pass** – Lights green when the HDMI source is active on the output in pass-through mode.

Ⓕ **USB ports** (future use)

Using the Standby Screen

Once the ShareLink Pro 1000 is connected and powered on, the standby screen appears on the display:

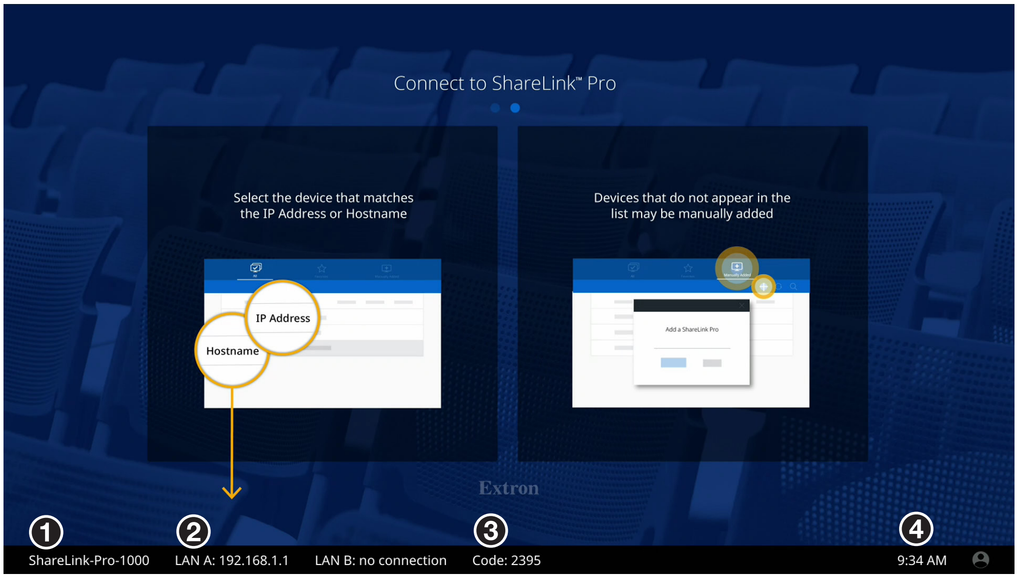


Figure 6. Standby Screen

❶ **Hostname** — Use this hostname to identify and access the ShareLink Pro 1000 during connection.

❷ **IP Address A and IP Address B** — Use this IP address to identify and access the ShareLink Pro 1000 during connection and to download the ShareLink Pro client software.

❸ **Login code** — Use this code to connect to the ShareLink Pro 1000 using the software. This code can be customized or disabled via the Product Configuration Software (PCS).

❹ **Time** — The time format can be adjusted via PCS.

NOTE:
- The standby screen background can be changed using PCS. For best results, make sure the desired background image matches the display output resolution.
- The status bar at the bottom of the standby screen (showing the IP addresses, hostname, and login code) is overlaid over the background image. Ensure that this portion of the background image does not contain important information as it will be covered by the status bar.

Operation Using a Computer

Users can present from a Windows or Mac computer using the ShareLink Pro 1000 software. This section covers the following:

- **Connecting a Computer to the ShareLink Pro 1000**
- **Downloading the ShareLink Pro 1000 Software**
- **Using the ShareLink Pro 1000 Software**
- **Using the Embedded Web Page**

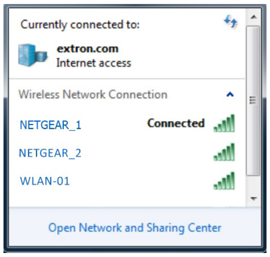
Connecting a Computer to the ShareLink Pro 1000

There are two ways to connect a computer to the ShareLink Pro 1000:

- Wirelessly, through an external wireless access point (WAP)
- Wired through a network device

See the following instructions for your preferred connection method.

Connecting through an External WAP



1. Ensure that the external WAP and the ShareLink Pro 1000 are wired to the same network.
2. Ensure that the ShareLink Pro 1000 is powered on and that wireless network capability is enabled on your computer.
3. Open the wireless networks on your PC and connect to the external WAP.

Connecting through a Network Hub, Switch, or Router

1. Ensure that your computer is connected to the network through a hub, switch, or router.
2. Connect the RJ-45 connector of the ShareLink Pro 1000 to the network device using an RJ-45 cable.

Downloading the ShareLink Pro 1000 Software

The ShareLink Pro 1000 software can be downloaded from the embedded web page or from the Extron website. Follow the instructions below for your preferred download method.

NOTE: For other details on the embedded web page, see **Using the Embedded Web Page** on page 15.

Downloading the software from the embedded web pages

1. Ensure that your computer is connected to the ShareLink Pro 1000 device, either through Wi-Fi or through a physical network connection. The computer and the ShareLink Pro 1000 must be on the same network.
2. Open your web browser and enter the ShareLink Pro IP address into the browser.

   NOTE: The IP address is shown on the standby screen (see **figure 6**, 1 on page 9).

The ShareLink Pro 1000 landing page opens on the browser:

Windows users: **Mac users**:

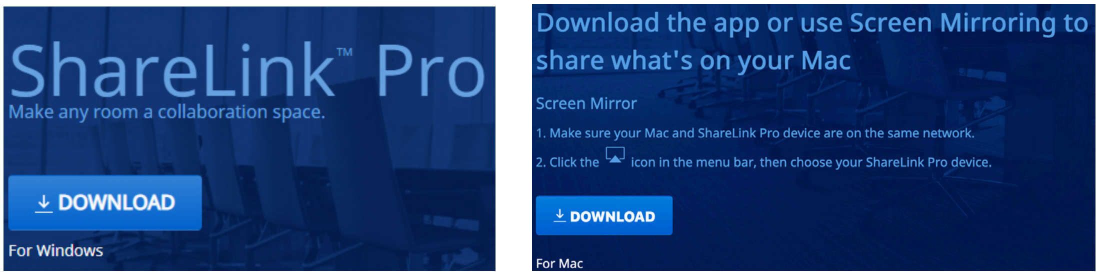


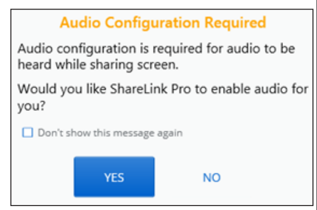
Figure 7. **ShareLink Pro Landing Page**

3. **Mac users**: Follow the instructions on the landing page or click **`Download`**.

   Windows users: Click **`Download`**, and submit any required information to run the executable (.exe) file.

> **NOTE: macOS users:** The .app software file is embedded in a .zip file. Some browsers automatically extract (unzip) the .app file. However, in some cases, the user may need to extract the file.

> **NOTE: Windows users:**
>
> - If audio configuration is required, the ShareLink Pro software displays the notification shown at right. Click **`Yes`** to allow the ShareLink Pro software to make the necessary audio configuration.
> - If an audio driver is required, the software will also ask for permission to install the Extron virtual audio driver. If you choose not to install the driver, you may do so later by clicking the information icon on the Share Screen (see **figure 11**, ❻, on page 13).



Downloading the software from the Extron website

1. On the Extron website, go to the **`Download`** tab and click **`Software`** (see figure 8, ❶).

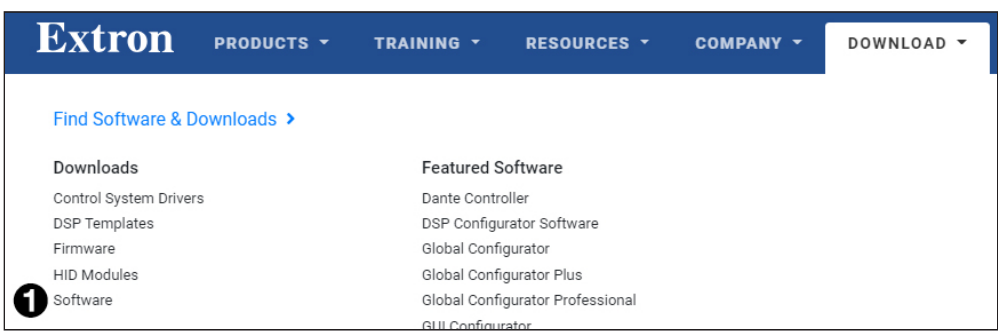


Figure 8. **Download Page on the Extron Website**

2. Click the **S** link (see figure 9, ❶).

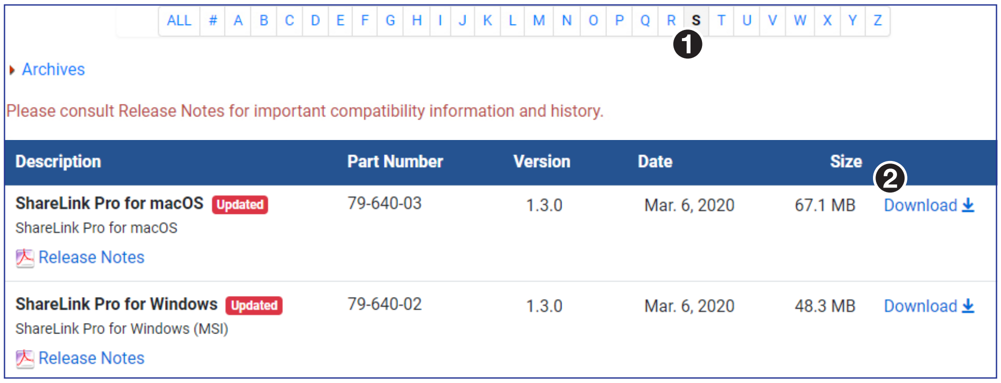


Figure 9. Software Download Link

3. Locate the ShareLink Pro software for your operating system and click `Download` (❷).
4. Submit any required information to start the download. Note where the file is saved.
5. Open the saved executable (.exe) file.
6. Follow the instructions that appear on the screen to install the program.

Using the ShareLink Pro 1000 Software

Connecting to the ShareLink Pro 1000

If the software is not yet connected to the ShareLink Pro 1000, follow these steps:

1. Open the ShareLink Pro software. The software searches for available devices on your network and lists them in the `All` panel.
2. Click the desired ShareLink Pro 1000 (see the standby screen for the device name).

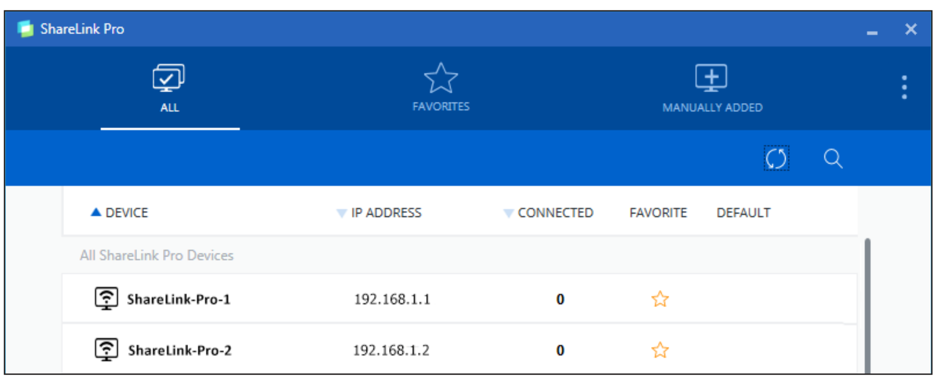


Figure 10. ShareLink Pro Devices List

> **NOTE:**
> - If your ShareLink unit is not listed, click the refresh button ( ).
> - If the device is still not listed, go to the `Manually Added` tab and select the add ( ) icon. Enter the hostname or IP address as instructed on the app screen.

3. When prompted, enter the login code and click `Connect`.

> **NOTE:** The login code is shown on the standby screen (see **figure 6** on page 9).

When connection is established, the `Share` screen appears (see **figure 11** on the next page).

Presenting content

Use the **`Share`** screen to present your computer screen on the display device.

NOTE:
- Up to four presenters can stream content simultaneously. The ShareLink Pro 1000 automatically arranges the incoming presentations side by side on the display.
- A moderator can control the content presented by participants during a session (see **Moderating a Presentation** on the next page for details).

Figure 11. Share Screen

❶ **Mirror Screen** — Click this button to automatically mirror your computer screen on the display device. Select **`Stop Presenting`** to stop the presentation.

❷ **Mirror Application** — Click this button to open a sub-menu, showing all the programs and applications currently running on your computer. Select a program or application to automatically mirror it on the display.

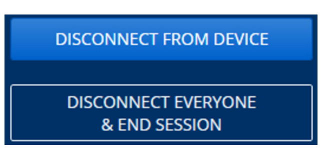
❸ **Share** — Click this button to navigate to a specific video or image file on your computer. Double-click the video or image file to present it on the display.

NOTE:
The following file formats are supported:
- **Image**: BMP, JPG, PNG, TIFF
- **Video**: 264, AVI, M2T, M2TS, M4V, MKV, MOV, MP4, MPG, SDP, TS

The following codecs are supported:
- **Video**: MPEG-2, MPEG4/XviD (Simple Profile), H.264 (Baseline, Main, High, Stereo SEI Profile), VC1 (Simple, Main, Advanced Profile), WEBM VP8, MJPEG VP8 (Mirror Screen feature only)
- **Audio**: AAC-LC (MPEG-2 part 7, MPEG-4 part 3, sub part 4 and part 14, MP4 mono or stereo audio, max 384 kbps)
PCM mono or stereo 16 kHz to 96 kHz, 16-bit to 32-bit
OPUS (Mirror Screen feature only)

❹ **View** — Click this tab to view the ongoing presentation on your screen.

❺ **End** — Click this tab to see these disconnection options:



- **`Disconnect from Device`** - Click this button to disconnect from the ShareLink Pro.
- **`Disconnect Everyone & End Session`** - Click this button to clear all shared content from the output and disconnect all users from the ShareLink Pro.

❻ **Information** — *This button is only shown if an appropriate audio driver has not been installed.* Click this button and follow the on-screen instructions to install the driver.

Moderating a Presentation

The ShareLink Pro 1000 default collaboration mode ("Run-Time" mode) allows a user to display content as a participant, or to control the presentation as a moderator. In this mode, everyone in the session starts as a participant, but a user can take over as moderator.

A moderator can accept or decline participants, stop shared content, and remove active participants from a session. Follow the instructions below to moderate a presentation.

NOTE: The following instructions apply only when the ShareLink Pro 1000 is set to Run-Time mode (the default setting). Run-Time mode can be disabled via PCS, and changed to either of the following modes:

- Full Control - No limitations to controls or sharing
- Moderate - Moderator manages access and sharing

1. Open the software menu by clicking the menu icon (see ❶ below), and click **`Moderate`** (❷).

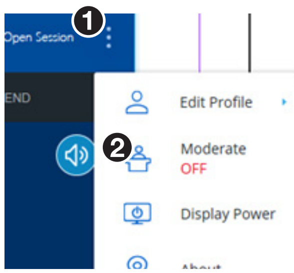


Figure 12. ShareLink Pro Software Menu

2. Click **`Enable Moderate`**. The `Manage` page appears on the software menu as shown below:

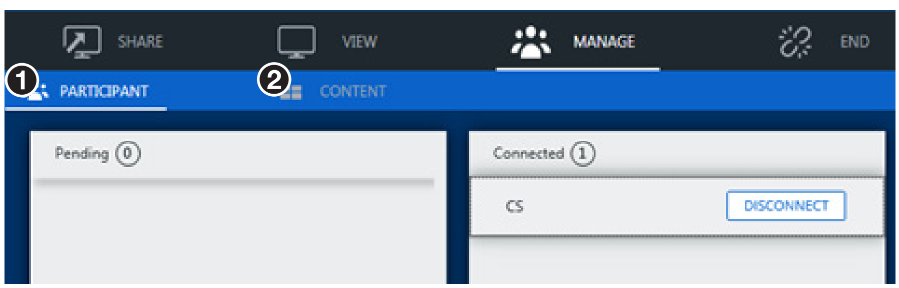


Figure 13. ShareLink Pro - Manage Page

❶ **`Participant tab`** — Use this section to manage session participants as follows:

- `Pending` (left) panel: Accept or decline pending participants listed in this panel.
- `Connected` (right) panel: Disconnect current participants by clicking **`Disconnect`** next to the applicable participant.

❷ **`Content tab`** — Use this section to stop the current presentation or content.

Using the Embedded Web Page

Viewing a Presentation using WebView

The WebView feature, available in the embedded web page, allows audience members to view a presentation on their computers or mobile devices through a web browser. The presenter may broadcast the presentation so that the image shown on the display can be viewed remotely.

NOTE: This feature can be disabled using PCS, or a password requirement can be set for using this feature.

Audience members must follow these steps to watch the presentation:

1. Open a web browser on the computer or mobile device, and enter the IP address of the ShareLink Pro 1000. The landing page opens as shown below.
2. Click **`View in WebView`** (see ❶, below). The presentation page opens in the web browser.

   NOTE: If necessary, enable WebView in the ShareLink Pro software (see ❹ on page 21).

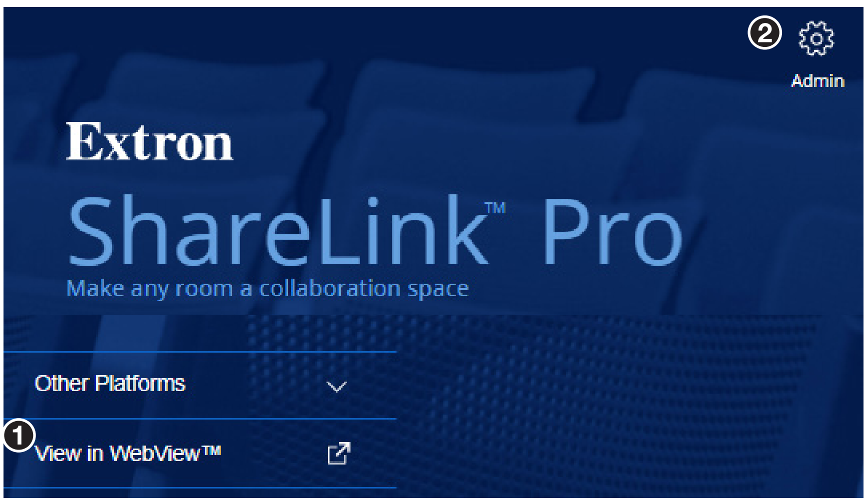


Figure 14. ShareLink Pro Landing Page

Using the Admin Page

To access the ShareLink Pro admin pages,

1. Click the settings icon on the top right corner (see figure 14, ❷ above).
2. Enter the user name and password, and click **`Login`**.
   - Default user name: `admin`
   - Default password: `extron`

NOTE: The factory configured password for this device has been set to the device serial number. Passwords are case sensitive. Performing a **Reset to Factory Defaults** (see page 7) sets the password to `extron`.

The admin page contains the following panels:

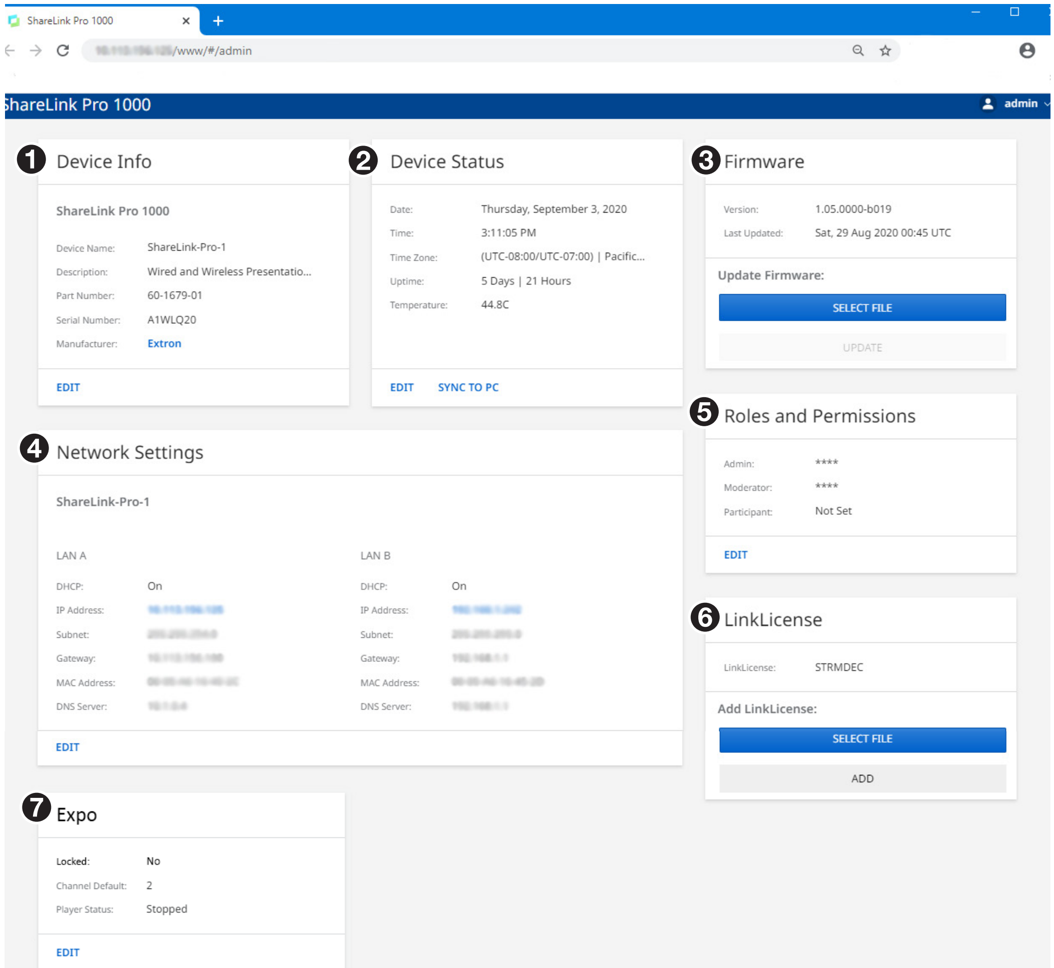


Figure 15. ShareLink Pro 1000 Admin Page

1. **Device Info**
2. **Device Status**
3. **Firmware**
4. **Network Settings**
5. **Roles and Permissions** (see next page)
6. **LinkLicense** (see next page)
7. **Expo** (see next page)

1. **Device Info** — Use this panel to change the device hostname. Click **`Edit`** to enter a new hostname, and click **`Save`**.

2. **Device Status** — Click **`Edit`** to change the device time, date, and time zone. Click **`Sync to PC`** to automatically sync the device to the PC timezone.

3. **Firmware** — Use this panel to update the device firmware:

   1. Click **`Select File`** to browse to the firmware file.

      NOTE: The latest firmware file can be downloaded from the Extron website (see Firmware Download on page 52).

   2. Select the firmware file, and click **`Update`**.

4. **Network Settings** — Click **`Edit`** to change the device name and the network settings (IP address, subnet, and gateway IP address) for the LAN port.

5 Roles and Permissions — Click **`Edit`** to change the following passwords:

> **NOTE:** The factory configured password for all accounts have been set to the device serial number. Passwords are case sensitive. Performing a **Reset to Factory Defaults** (see page 7) sets the password to the defaults shown below.

- **Admin password** (default: **`extron`**) — This password is used for accessing the admin page.
- **Moderator password** (default: **`moderate`**) — This password is required to moderate a session.
- **Participant password** (default: *no password*) — This password is required to participate in a session.

Enter and confirm the new password, and click **`Save`**.

6 LinkLicense — Use this panel to upload the LinkLicense file to the ShareLink Pro 1000:

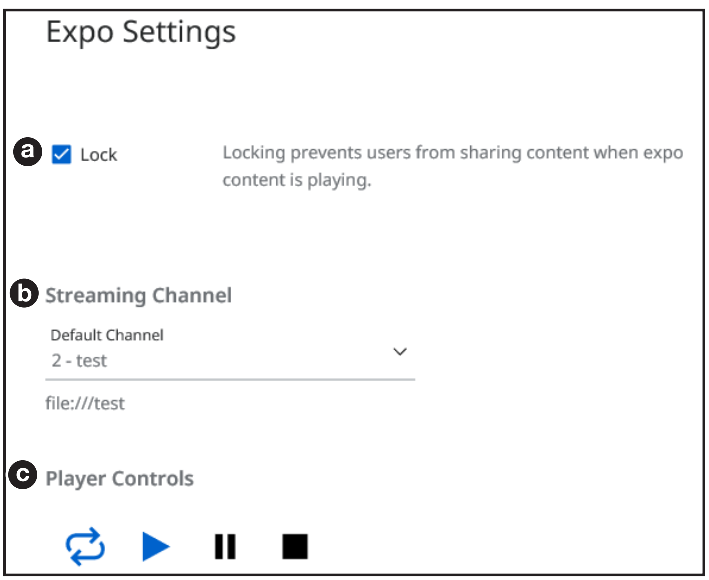
> **NOTE:** A LinkLicense unlocks the Expo feature (streaming capability) for the ShareLink Pro 1000. To redeem (activate) a LinkLicense, go to **www.extron.com/llredeem** and follow the online instructions.

If a LinkLicense has been purchased, upload the file as follows:

a. Click **`Select File`** and browse to the LinkLicense (`.ell`) file on your computer.

b. Click **`Add`** to upload the file.

7 Expo — Click **`Edit`** to open the `Expo Settings` page:



a Lock — Click this checkbox to enable the Lock feature. This prevents users from sharing content when Expo content is playing.

b Streaming Channel — Click this dropdown menu to select the channel that you wish to present.

> **NOTE:** Streaming channels can be added or removed using Extron PCS (see the *ShareLink Pro Help File* in PCS).

c Player Controls — Use this panel to play, pause, and stop Expo content. Click the repeat ( ) button to automatically replay expo content.

Operation Using a Mobile Device

The **ShareLink Pro 1000 Mobile App** is supported by Android and Apple mobile devices, including iPad and iPhone. It allows users to present photos, web pages, device screenshots, and the device camera (see instructions below).

The ShareLink Pro 1000 also supports screen mirroring for Apple devices.

This section covers the following:

- **Downloading the ShareLink Pro 1000 App**
- **Using the ShareLink Pro 1000 App**
- **Using iOS Screen Mirroring**

Downloading ShareLink Pro 1000 App

Follow these steps to download the ShareLink Pro 1000 mobile app:

1. Ensure that the ShareLink Pro is powered and connected to the network.
2. Open the Wi-Fi networks list on your device and connect to an external WAP that is on the same network as the ShareLink Pro. Ensure that the WAP and the ShareLink Pro are wired to the same network.
3. Navigate to the ShareLink Pro landing page (see figure 16) by entering the device IP address into a browser on the mobile device (see the standby screen for the ShareLink Pro IP address).

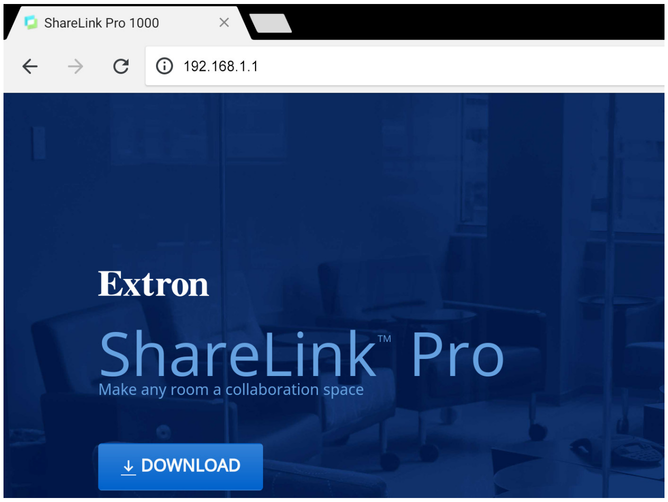


Figure 16. ShareLink Pro Landing Page

4. Select **`Download`**. The ShareLink Pro app download page opens on the app store for your device.
5. Install the app on your mobile device.

Using the ShareLink Pro 1000 App

Connecting to a ShareLink Pro Unit

Once the ShareLink Pro 1000 app is installed on the device, follow these steps to connect to the ShareLink Pro:

1. Ensure that the ShareLink Pro is powered and connected to the network.
2. Open the Wi-Fi networks list on your device and connect to an external WAP that is on the same network as the ShareLink Pro. Ensure that the WAP and the ShareLink Pro are wired to the same network.
3. Open the ShareLink Pro 1000 app and select the desired unit from the list (see the standby screen to verify the device hostname and IP address).

NOTE: To refresh, drag down and release the list. If the unit is not listed, go to the `Manually Added` tab and select the add (⊞) icon. Enter the hostname or IP address as instructed on the app screen.

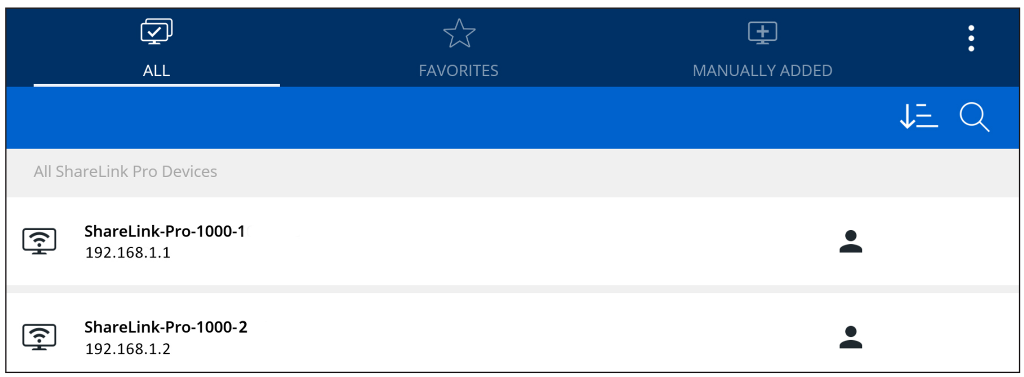


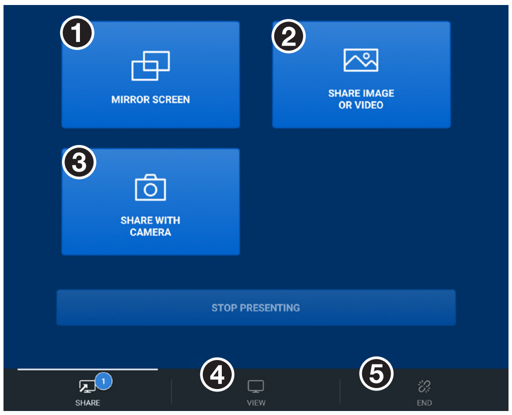
Figure 17. ShareLink Pro Devices List

4. When prompted, enter the login code shown on the standby screen.
5. When connection is established, the app shows the `Share` screen (see the next page).

Presenting Content

The **Share** screen contains the following options to present content:

NOTE:
- Up to four presenters can stream content simultaneously. The ShareLink Pro automatically arranges the incoming presentations side by side on the display.
- A moderator can control the content presented by participants during a session (see **Moderating a Presentation** on page 14 for details).



❶ **Mirror Screen** – Click this button to automatically mirror your device screen on the display device. Select **Stop Presenting** to stop the presentation.

NOTE: For iOS devices, follow the on-screen instructions for mirroring the device screen, or see **Using iOS Screen Mirroring** on page 22.

❷ **Share Image or Video** – Click this button to browse to a specific video or image file on your device. Select the video or image file to present it on the display.

❸ **Share with Camera** – Click this button to share an image taken by the device camera.

NOTE: For iOS devices, a **Share Website** option is also available to share the device web browser.

❹ **View** – Click this tab to view the ongoing presentation on your screen.

❺ **End** – Click this tab to disconnect from the ShareLink Pro 1000.

App Menu

Press the menu icon (⋮) on the top right corner to see the following options:

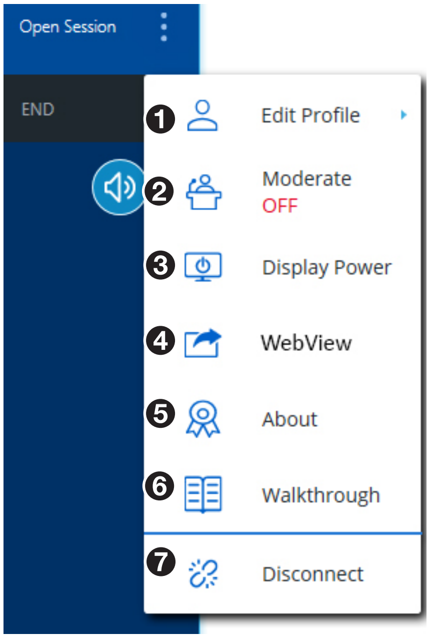


Figure 18. App Menu

❶ **Edit Profile** — Select this option to edit your Username and Login Settings.

❷ **Moderate** — Select this option to enable Moderator mode (see **Moderating a Presentation** on page 14 for details).

❸ **Display Power** — This option opens a page with two buttons (**ON** and **OFF**). If the display connected to the output of the ShareLink Pro supports CEC, the **ON** and **OFF** buttons may be used to send out unidirectional Power ON and Power OFF commands to the display via CEC.

❹ **WebView** — Select this option to enable or disable WebView.

❺ **About** — Select this option to see the app version number, Licenses, Terms of Use, and Privacy Policy.

❻ **Walkthrough** — Select this option to see the introductory walkthrough that is shown when the app is first installed.

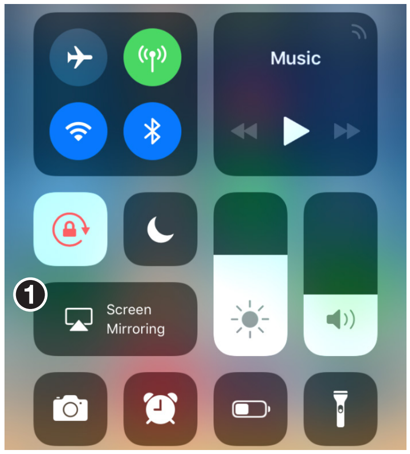
❼ **Disconnect** — Select this option to disconnect from the ShareLink Pro.

Using iOS Screen Mirroring

This feature allows iOS device screens to be mirrored onto the display.

1. Ensure that the ShareLink Pro 1000 is powered and connected to the network.
2. Connect your device to the ShareLink Pro 1000 through Wi-Fi (see step 2 of **Connecting to a ShareLink Pro Unit** on page 19).
3. Open the iOS device control center and select **`Screen Mirroring`** (see ❶, below).

   NOTE: Options may look different depending on the iOS version.

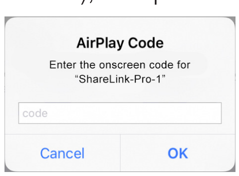


The discovered devices list appears as shown at right.

4. Select your ShareLink Pro device from the discovered devices list.

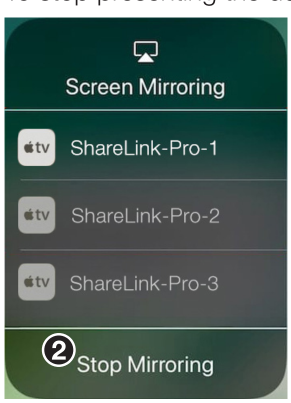
   NOTE: If your ShareLink Pro device is not listed because it is on a different subnet, follow the instructions on the next page.

If prompted for a code (as shown below), enter the ShareLink Pro login code (as shown on the standby screen), and press **`OK`**.



The device screen is immediately mirrored onto the display.

To stop presenting the device screen, select **`Stop Mirroring`** (see ❷, below).



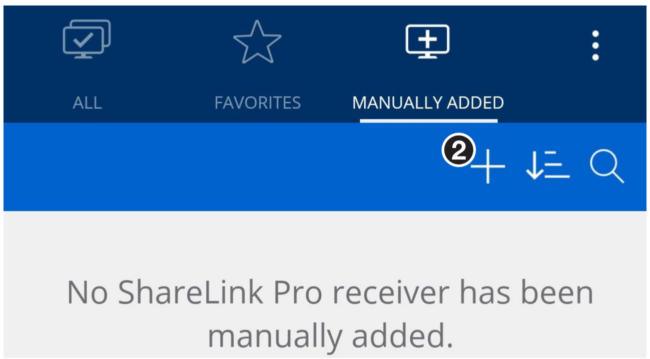
Discovering a ShareLink Pro device on a different subnet for Apple Screen Mirroring

The ShareLink Pro for iOS mobile app automatically discovers ShareLink Pro devices that are on the same subnet as the iOS device.

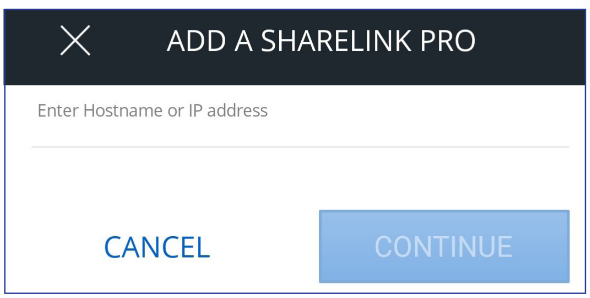
If your ShareLink Pro device is not on the same subnet, follow these steps to discover the device using the app.

> **NOTE:** The same steps apply to macOS devices using the ShareLink Pro macOS software.

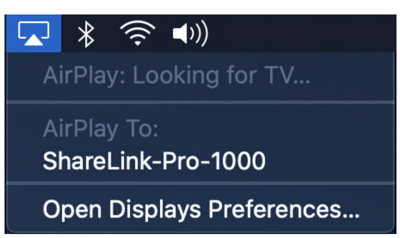
1. Open ShareLink Pro for iOS mobile app.
2. Go to the `Manually Added` tab (shown below) and select the + icon (❷).



3. Enter the IP address or hostname of the ShareLink Pro device that is outside the local subnet.



4. Enter password if prompted.
5. Enter confirmation if prompted.
6. Once the connection has been successfully established and the main sharing page is shown:
   - **For iOS devices**: Go to the Control Center and navigate to the Screen Mirroring device discovery list (see step 3 on the previous page).
   - **For macOS devices**: Click the AirPlay button in the menu bar as shown below.



7. Select the ShareLink Pro device from the device discovery list (see step 4 on the previous page).

SIS Configuration and Control

This section contains SIS communication details and SIS commands and responses for the ShareLink Pro 1000. Topics in this section include:

- **Host Device Connection**
- **SIS Overview**
- **Command and Response Tables for SIS Commands**

Host Device Connection

Use a connected computer running an SSH client such as PuTTY to send and receive SIS commands and responses.

Use port 22023 for connection to the ShareLink Pro when using a control system. Enter the admin username and password (default password: `extron`) when prompted.

> **NOTE:** The factory configured password for all accounts have been set to the device serial number (passwords are case sensitive). Performing a **Reset to Factory Defaults** (see page 7) sets the password to `extron`.

SIS Overview

Host and Device Communication

SIS commands consist of one or more characters per field. No special characters are required to begin or end a command sequence. When the ShareLink Pro 1000 determines that a command is valid, it executes the command and sends a response to the host device. All responses from the device to the host end with a carriage return and a line feed (CR/LF = ↵), which signals the end of the response character string. A string is one or more characters.

Error Responses

When the ShareLink Pro 1000 receives an SIS command and determines that it is valid, it performs the command and sends the corresponding response to the host device. If the command is determined invalid or contains invalid parameters, the device returns an error response to the host. The error response codes are:

`E10` = Invalid command
`E11` = Invalid preset number
`E13` = Invalid parameter
`E14` = Not valid for this configuration
`E17` = Invalid command for signal type
`E22` = Busy
`E24` = Privilege violation

Command and Response Tables Overview

The command and response tables for SIS commands list the commands the device recognizes as valid, the responses returned to the host, a description of the command function or results of executing the command, and some examples of commands in ASCII.

NOTE: Upper and lowercase text can be used interchangeably unless otherwise stated.

Symbol Definitions

The table below shows the hexadecimal equivalent of ASCII characters used in the command and response tables. Hexadecimal values include two digits. The following symbols are commonly used throughout the command and response tables.

ASCII and Hexadecimal Conversion Table

First Hexadecimal Digit	Second Hexadecimal Digit															
	0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**A**	**B**	**C**	**D**	**E**	**F**
0											LF			←		
1												Esc				
2	●	!	“	#	$	%	&	‘	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	z	y	z	{	|	}	~	

↵ = Carriage return and line feed (LF)

← = Carriage return with no line feed

| = Pipe (can be used interchangeably with the ← character).

● = Space

Esc = Escape key

W = Can be used interchangeably with the Esc character.

X1 = Input window status — `0` = HDMI window not displayed `1` = HDMI window displayed

X2 = Input signal — `0` = No input signal detected `1` = Input signal detected

X3 = HDMI Input Signal Detected Rate (Ex: `1920x1080@60.00Hz`); no signal = `0000x0000@00.00Hz`

X4 = Video format — `0` = No signal, `1` = DVI, `2` = HDMI

X5 = EDID table slot

X6 = Native resolution & refresh rate — Ex: `1920x1080@60.00Hz`

X7 = EDID data in HEX

X8 = Output rate (see table on page 30)

X9 = Output sync mode
- `0` = Sync always active (default)
- `1` = No signal present timer, disable output
- `2` = Inactivity timer, disable output

X10 = Time out duration
- `0-999999` (1 second increments)
- `1000000` = Never times out
- Default = 30 seconds

X11 = Output sync mode status
- `0` = Active input detected; timer not running
- `1` = No active input; timer is running; output sync still active
- `2` = No active input; timer has expired; output sync disabled

X12 = Color bit depth	0 = Auto mode (default) 1 = Force 8-bit
X13 = Output video format	0 = Auto (default) 1 = DVI RGB 4:4:4 Full 2 = HDMI RGB 4:4:4 Full 3 = HDMI RGB 4:4:4 Limited 4 = HDMI YUV 4:4:4 Limited 5 = HDMI YUV 4:2:2 Limited 6 = HDMI YUV 4:2:0 Limited
X14 = HDCP status	0 = Not connected 1 = Not HDCP Encrypted 2 = HDCP Encrypted
X15 = HDCP authorized status	0 = HDCP authorize off 1 = HDCP authorize on (default)
X16 = HDCP mode	0 = Follow input (default) 1 = Always encrypt 2 = Follow input with continuous DVI trials 3 = Always encrypt with continuous DVI trials 4 = Disable authentication
X17 = HDCP notification	0 = Black screen 1 = Green screen (default)
X18 = Video mute status	0 = Unmute (default) 1 = Mute video 2 = Mute sync
X19 = Audio mute status	0 = Unmute (default) 1 = mute
X21 = Input type	1 = HDMI 2 = Decoder (for all decoder streams) 3 = Airplay streams 4 = Uploaded image/video content
X22 = Input aspect ratio setting	1 = Fill 2 = Follow (default)
X23 = Volume adjustment range	-100 to 0 = (default is -30dB)
X25 = Contact closure input	1 = Contact closure input 1 2 = Contact closure input 2 3 = Contact closure input 3 4 = Contact closure input 4
X26 = Tally output	1 = Tally output 1 2 = Tally output 2 3 = Tally output 3 4 = Tally output 4
X27 = Contact/tally state	0 = Off (open) 1 = On (closed)
X28 = Executive mode	0 = Off (default) 1 = On
X30 = Unit name	Alpha-numeric up to 63 characters. No blanks, spaces, or special characters except hyphen (-).
X31 = Default unit name	Alpha-numeric name. ShareLink-Pro-xx-xx-xx (where xx-xx-xx is the last six digits of the unique MAC address).
X32 = Firmware version (Ex: 1.00)	
X33 = Firmware version and build version (Ex: 1.00.0010)	
X34 = Internal temperature (2 digits, °C)	
X35 = Verbose response mode	0 = Clear (none) (default), 1 = Verbose mode, 2 = Tagged responses, 3 = Verbose + tagged responses
X36 = Layout mode	1 = Layout group #1 (default) 2 = Layout group #2 3 = Layout group #3 4 = Layout group #4 5 = Layout group #5
X37 = Current layout number (1-15)	
X38 = HDMI input mode	0 = Disabled 1 = Enabled (default)
X39 = Apple screen mirroring mode	0 = Disabled 1 = Enabled (default)
X391 = Device discovery proxy service	0 = Disabled (default) 1 = Enabled
X40 = HDMI controls setting	0 = Disabled 1 = Enabled (default)
X401 = HDMI client app controls	0 = Hide 1 = Show (default)
X41 = Client app controls label	UTF-8 characters permitted, maximum of 16 characters

X42 = Access mode	1 = Full control 2 = Moderated 3 = Run-time (default)
X43 = Run time access mode	1 = Full control 2 = Moderated
X44 = Confirmation code mode	0 = None 1 = Fixed 2 = Random (default)
X45 = Login key code	Four digit value (default = 0000)
X46 = Random code interval	Four digit value (1 to 9998) in minutes Default = 9999
X47 = WebView password option	0 = No access (disabled) 1 = No prompt for password (default) 2 = Custom password 3 = Follow confirmation code
X48 = Status bar mode	0 = Never shown 1 = Always shown (default) 2 = Not shown only when one content is shared 3 = Not shown when one or more content is shared
X49 = Label	1 = Hostname 2 = IP address A 3 = IP address B, 4 = Login code 5 = Custom message 6 = Dynamic notifications/messages 7 = Clock
X50 = Label value (max 128 characters)	
X51 = Label visibility setting	0 = Hide 1 = Show (default)
X52 = Language	1 = English (default) 2 = French 3 = German 4 = Spanish 5 = Chinese 6 = Japanese
X53 = Still image transition effect	0 = Cut (default) 1 = Fade
X54 = Still image transition time	500 to 3000ms increments (default = 1000ms)
X55 = Time between still image sequences	1000 to 60000ms in 1ms increments (default = 10000ms)
X56 = Gallery setting	1 = Standby image gallery 2 = Connected image gallery
X57 = Port mode	0 = Disabled, 1 = Built-in control (default) 2 = RS-232 Over LAN Pass-Thru
X58 = Baud rate	1200, 1800, 2400, 3600, 4800, 7200, 9600, 14400, 19200, 28800, 38400, 57600, 115200 (9600 = default)
X59 = Parity	**O**dd, **E**ven, **N**one, **M**ark, **S**pace (only the first letter is required; **N**one = default)
X60 = Data bits	7 or 8 (8 = default)
X61 = Stop bits	1 or 2 (1 = default)
X62 = Port timeout	1 (10 seconds) to 65000 (650,000 seconds) Default is 30 (300 seconds)
X63 = Start Point for UART port	(Default = 2001)
X64 = Connection ID (10 digits)	

X65 = User name (apha-numeric text)

X66 = Stream ID (10 digits) (if connection is not streaming, this will be empty)

X67 = Content type (example: HDMI, App Window, AirPlay, etc)

X68 = Connection approved (0 = pending, 1 = approved)

X69 = X: x position of the window as a percentage of the screen width times 100

X70 = Y: y position of the window as a percentage of the screen height times 100

X71 = W: width of the window as a percentage of the screen width times 100

X72 = H: height of the window as a percentage of the screen height times 100

X73 = Connection age (in seconds)

X74 = Total number of users connected (integer – maximum 64)

X75 = Video port ID (7 digits)

X76 = Echo status — `0` = Disabled `1` = Enabled (default)

X77 = Color described as RGB Hex triplets (RRGGBB)

X78 = mDNS hostname (alpha-numeric up to 63 characters; no blanks, spaces, or special characters except hyphen (-).

> **NOTE:** The first character must be an alphabetical character, and the last character must not be a hyphen.

X79 = Status bar pop-up — `0` = Disabled `1` = Enabled (default)

X791 = Status bar pop-up timer — 1s-300s in 1s increments (default = 20s)

X80 = Stream mode — `1` = Audio, `2` = Video, `3` = Audio and video (default)

X81 = Source audio sample rate — `0` = Reserved, `1` = Reserved, `2` = 44.1 kHz, `3` = 48 kHz

X82 = Playback channel (always 1 for single channel devices)

X83 = Playback speed — `1` = normal speed

X84 = Player state — `0` = Stop, `1` = Play, `2` = Pause

X85 = Loop play status — `0` = Off/Disable, `1` = On/Enable

X86 = Track number (`1` to maximum tracks currently defined)

X87 = Timecode (`HH:MM:SS:DD`, where `DD` is decimal seconds up to nine digits)

X88 = Time (in seconds)

X89 = File path

X90 = Media URI (Uniform Resource Identifier). The prefix “`file:///`” is optional.

X91 = Preset number, and Media URI — Example: `[{“preset”:1, “uri”:file:///clips/clip1.mp4, “name”:“Chan1”}, …]`

X93 = Channel number — `1 - 999`

X104 = Expo feature — `0` = Disabled (default), `1` = Continuous, `2` = Temporary

X105 = Player state — `1` = Standby `2` = Connected `3` = Expo `4` = Expo Standby `5` = Sharing

X106 = Service (network path of server) — **CIFS**: “`\\\\myServer\\myShare\\myDir`” **NFS**: “`myServer:/myDir`”

X107 = Mountpoint

X108 = Username

X109 = Password

X110 = Options

X111 = Boolean for `reconnect`

X112 = Boolean for `extron_share`

X113 = Recursion (optional) — `0` = Disabled, `1` = Enabled (default)

X114 = Directory path (optional)

X115 = Network interface — `1` = LAN A `2` = LAN B

X116 = Initial play threshold (sec)

X117 = Rebuffering play threshold (sec)

Command and Response Tables for SIS Commands

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Video input selection			
Select HDMI window input	Esc 1PIPS←	Pips1↵	
De-select HDMI window input	Esc 0PIPS←	Pips0↵	
View HDMI window input	Esc PIPS←	X1↵	
Select HDMI pass-through input	1!	In1↵	
De-select HDMI pass-through input	0!	In0↵	
View HDMI pass-through input	!	X1↵	
		InX1↵	(Verbose mode 2/3)
Query HDMI input signal detected rate	Esc 1LS←	X3↵	
		In01•X3↵	(Verbose mode 2/3)
View detected format	1*\	X4↵	View video format on HDMI input
		Vtyp1*X4↵	(Verbose mode 2/3)
Query HDMI input signal presence	Esc 0LS←	X2↵	View HDMI signal presence
Input EDID (HDMI)			
Import EDID (.bin) to input (store) slot	Esc IX5,*<filename>* EDID←	EdidIX5↵	Import EDID from *<filename>* to input slot X5
Export EDID (.bin) to PC	Esc EX5,*<filename>* EDID←	EdidEX5↵	Export EDID from EDID table slot X5 to *<filename>*
View EDID native resolution	Esc NX5EDID←	X6↵	View native resolution and refresh rate of EDID from EDID table slot X5. (Ex: `1920x1080@60.00Hz`)
		EdidNX5*X6↵	(Verbose mode 2/3)
View/Read EDID of input in Hex format	Esc R1EDID←	X7↵	
		EdidR1*X7↵	(Verbose mode 2/3)

KEY:

- X1 = Input window status — `0` = HDMI window not displayed   1 = HDMI window displayed
- X2 = Input signal — `0` = No input signal detected   `1` = Input signal detected
- X3 = HDMI Input Signal Detected Rate — (Ex: `1920x1080@60.00Hz`); no signal = `0x0@0Hz`
- X4 = Video format — `0` = No signal, `1` = DVI, `2` = HDMI
- X5 = EDID table slot
- X6 = Native resolution & refresh rate — Ex: 1920x1080@60.00Hz
- X7 = EDID data in HEX

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Output Rate			
Set output rate	Esc 1*X8RATE←	Rate1*X8↵	Set output rate
View output rate	Esc 1RATE←	X8↵	View output rate
		Rate1*X8↵	(Verbose mode 2/3)

KEY:

X8 = Output rate (see the table below; default = 45).

Resolution	23.98 Hz	24 Hz	25 Hz	29.97 Hz	30 Hz	50 Hz	59.94 Hz	60 Hz
640x480								10
800x600								11
1024x768								12
1280x768								13
1280x800								14
1280x1024								15
1360x768								16
1366x768								17
1440x900								18
1400x1050								19
1600x900								20
1680x1050								21
1600x1200								22
1920x1200								23
480p							24	25
576p						26		
720p			29	30	31	32	33	34
1080i						35	36	37
1080p	38	39	40	41	42	43	44	45*
2K (2048x1080)	46	47	48	49	50	51	52	53
2048x1200								54
2048x1536								55
2560x1080								56
2560x1440								57
2560x1600								58
4K (3840x2160)	59	60	61	62	63			
4096x2160	69	70	71	72	73			

* = Default

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Screen saver			
Set output sync rate	Esc M1*X9SSAV←	SsavM1*X9↵	Sets output sync mode
View output sync mode	Esc M1SSAV←	X9↵ SsavM1*X9↵	View output sync mode (Verbose mode 2/3)
Set time out duration	Esc T1*X10SSAV←	SsavT1*X10↵	Sets the time out duration to X10 seconds
View time out duration	Esc T1SSAV←	X10↵ SsavT1*X10↵	View current time out duration X10 (Verbose mode 2/3)
View screen saver status	Esc S1SSAV←	X11↵ SsavS1*X11↵	View the current screen saver status status X11 (Verbose mode 2/3)
Color bit depth			
Set video bit depth	Esc V1*X12BITD←	BitdV1*X12↵	
Query video bit depth	Esc V1BITD←	X12↵ BitdV1*X12↵	View color bit depth (Verbose mode 2/3)
Video output format			
Set format	Esc 1*X13VTPO←	Vtpo1*X13↵	Sets output video format
View setting	Esc 1VTPO←	X13↵ Vtpo1*X13↵	View output video format (Verbose mode 2/3)

KEY:

X9 = Output sync mode
0 = Sync always active (default)
1 = No signal present timer, disable output
2 = Inactivity timer, disable output

X10 = Time out duration
0-999999 (1 second increments)
1000000 = Never times out
Default = 30 seconds

X11 = Screen saver status
0 = Active input detected; timer not running
1 = No active input; timer is running; output sync still active
2 = No active input; timer has expired; output sync disabled

X12 = Color bit depth
0 = Auto mode (default)   1 = Force 8-bit

X13 = Output video format
0 = Auto (default)   1 = DVI RGB 4:4:4 Full
2 = HDMI RGB 4:4:4 Full   3 = HDMI RGB 4:4:4 Limited
4 = HDMI YUV 4:4:4 Limited   5 = HDMI YUV 4:2:2 Limited
6 = HDMI YUV 4:2:0 Limited

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
HDCP			
Set HDCP mode	Esc S1*X16HDCP←	HdcpS1*X16↵	
Query HDCP mode	Esc S1HDCP←	X16↵	
		HdcpS1*X16↵	(Verbose mode 2/3)
HDCP Authorized Device On	Esc E1*1HDCP←	HdcpE1*1↵	HDMI input reports as an HDCP Authorized Device
HDCP Authorized Device Off	Esc E1*0HDCP←	HdcpE1*0↵	HDMI input reports as a non-HDCP Authorized Device
Query HDCP Authorized Device Status	Esc E1HDCP←	X15↵	View HDCP authorized setting
		HdcpE1*X15↵	(Verbose mode 2/3)
View input HDCP status	Esc I1HDCP←	X14↵	View HDCP status on HDMI input
		HdcpI1*X14↵	(Verbose mode 2/3)
View output HDCP status	Esc O1HDCP←	X14↵	View HDCP status on HDMI output
		HdcpO1*X14↵	(Verbose mode 2/3)
HDCP notification			
Enable HDCP notification	Esc N1*1HDCP←	HdcpN1*1↵	
Disable HDCP notification	Esc N1*0HDCP←	HdcpN1*0↵	
Query HDCP notification	Esc N1HDCP←	X17↵	View current HDCP notification setting
		HdcpN1*X17↵	(Verbose mode 2/3)
Video Mute			
Mute video to black	1*1B/b	Vmt1*1↵	Mute video on HDMI output
Mute sync and video	1*2B/b	Vmt1*2↵	Mute sync and video on HDMI output
Video unmute	1*0B/b	Vmt1*0↵	Unmute video on HDMI output
View video mute	1B/b	X18↵	View current video mute status
		Vmt1*X18↵	(Verbose mode 2/3)

KEY:

X14 = HDCP status	0 = Not connected 1 = Not HDCP Encrypted 2 = HDCP Encrypted
X15 = HDCP authorized status	0 = HDCP authorize off 1 = HDCP authorize on (default)
X16 = HDCP mode	0 = Follow input (default) 1 = Always encrypt 2 = Follow input with continuous DVI trials 3 = Always encrypt with continuous DVI trials 4 = Disable authentication
X17 = HDCP notification	0 = Black screen 1 = Green screen (default)
X18 = Video mute status	0 = Unmute (default) 1 = Mute video 2 = Mute sync

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Output Audio Mute			
Mute audio output	`1*1Z/z`	`Amt1*1↵`	Mute audio outputs (mutes both HDMI and Analog audio)
Unmute audio output	`1*0Z/z`	`Amt1*0↵`	Unmute audio outputs (unmutes both HDMI & Analog audio)
View audio output mute	`1Z/z`	X19↵	View current audio mute status
		`Amt1*`X19↵	(Verbose mode 2/3)
Input aspect ratio (per input)			
Set to fill	Esc X21`*1ASPR`←	`Aspr`X21`*1`↵	Set input aspect ratio to FILL
Set to follow	Esc X21`*2ASPR`←	`Aspr`X21`*2`↵	Set input aspect ratio to FOLLOW
View aspect ratio setting	Esc X21`ASPR`←	X22↵	View input aspect ratio setting
		`Aspr`X21`*`X22↵	(Verbose mode 2/3)
Analog Audio Output Volume			
Specify volume	X23`V/v`	`Vol`X23↵	Set volume level. Response is 3 digits. Ex: **`Vol-050`**
Increment volume	`+V`	`Vol`X23↵	Volume level is X23
Decrement volume	`-V`	`Vol`X23↵	Volume level is X23
View volume	`V`	X23↵	View volume level

KEY:

X19 = Mute status	`0` = Unmute (default)	1 = mute
X21 = Input type	`1` = HDMI	
X22 = Input aspect ratio setting	`1` = Fill	`2` = Follow (default)
X23 = Volume adjustment range	**`-100`** to **`0`** = (default is **`-30dB`**)	

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Contact/Tally Ports			
Query contact closure state	Esc X25 CNTC←	X27↵	View contact X25 closure state
		Cntc X25 * X27↵	(Verbose mode 2/3)
Query all contact closure states	Esc CNTC←	X27•X27•X27•X27↵	View all contact closure states
		Cntc* X27•X27•X27•X27↵	(Verbose mode 2/3)
Set tally output	Esc X26 * X27 TALY←	Taly X26 * X27↵	Set tally output X26 state
Query tally output	Esc X26 TALY←	X27↵	View tally output X26 state
		Taly X26 * X27↵	(Verbose mode 2/3)
Set all tally outputs	Esc X27 *TALY←	Taly X27↵	Set all tally output states to X27
Query all tally outputs	Esc TALY←	X27•X27•X27•X27↵	View all tally output states
		Taly* X27•X27•X27•X27↵	(Verbose mode 2/3)
Front Panel Lockout			
Front panel executive mode	X28 X	Exe X28↵	Lock/unlock front panel
View lockout status	X/x	X28↵	
		Exe X28↵	(Verbose mode 2/3)

KEY:

X25 = Contact closure input	1 = Contact closure input 1	2 = Contact closure input 2
	3 = Contact closure input 3	4 = Contact closure input 4
X26 = Tally output	1 = Tally output 1	2 = Tally output 2
	3 = Tally output 3	4 = Tally output 4
X27 = Contact/tally state	0 = Off (open)	
	1 = On (closed)	
X28 = Executive mode	0 = Off (default)	
	1 = On	

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Unit name			
Set unit name/ hostname	Esc X30 CN←	Ipn•X30↵	
Set unit name/ hostname to default	Esc •CN←	Ipn•X31↵	Default = `ShareLink-Pro-xx-xx-xx` (where `xx-xx-xx` is the last six digits of the unique MAC address).
View unit name/ hostname	Esc CN←	X30↵ Ipn•X30↵	View unit name/hostname (Verbose mode 2/3)
Information			
Information request (View unit information)	I/i	Inf*Frq X2•HdcpI X14•HdcpO X14•Vmt X18•Amt X19•Vol X23↵ *Verbose mode 2/3:* Inf00*Frq X2•HdcpI X14•HdcpO X14•Vmt X18•Amt X19•Vol X23↵	
Query firmware version	Q/q	X32↵	View firmware version
Query firmware and build version	*Q/q	X33↵	View firmware version and build version
Query internal temperature	Esc 20STAT←	X34↵ 20Stat X34↵	Query internal temperature (Verbose mode 2/3)
Query model name	1I/i	<model name>↵ Inf01*<model name>↵	View model name (Verbose mode 2/3)
Query model description	2I/i	<model description>↵ Inf02*<model desc.>↵	View model description (Verbose mode 2/3)
Query part number	N/n	<part number>↵ Pno<part number>↵	View part number (Verbose mode 2/3)

KEY:

X2 = Input signal	`0` = No input signal detected `1` = Input signal detected
X14 = HDCP status	`0` = Not connected, `1` = Not HDCP Encrypted, `2` = HDCP Encrypted
X18 = Video mute status	`0` = Unmute (default), `1` = Mute video, `2` = Mute sync
X19 = Audio mute status	`0` = Unmute (default), `1` = Mute
X23 = Volume adjustment range	`-100` to `0` = (default is `-30dB`)
X30 = Unit name	Alpha-numeric up to 63 characters. No blanks, spaces, or special characters except hyphen (-).
X31 = Default unit name	Alpha-numeric name. `ShareLink-Pro-xx-xx-xx` (where `xx-xx-xx` is the last six digits of the unique MAC address).
X32 = Firmware version (Ex: `1.00`)	
X33 = Firmware version and build version (Ex: `1.00.0010`)	
X34 = Internal temperature (2 digits, °C)	

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Verbose mode			
Set verbose mode	Esc X35 CV ←	Vrb X35 ↵	
View verbose mode	Esc CV ←	X35 ↵	
Output display layout			
Set display layout mode	Esc X36 OMOD ←	Omod X36 * X37 ↵	
View display layout mode	Esc OMOD ←	X36 * X37 ↵	
		Omod X36 * X37 ↵	(Verbose mode 2/3)
Auto-default to HDMI input			
Set auto-default to HDMI input mode	Esc X38 AUSW ←	Ausw X38 ↵	
View auto-default to HDMI input mode	Esc AUSW ←	X38 ↵	
		Ausw X38 ↵	(Verbose mode 2/3)
Apple screen mirroring			
Set Apple Screen Mirroring	Esc A X29 * X39 SHAR ←	SharA X29 * X39 ↵	
View Apple Screen Mirroring	Esc A X29 SHAR ←	X39 ↵	
		SharA X29 * X39 ↵	(Verbose mode 2/3)
Set Apple Screen Mirroring Device Discovery Proxy Service	Esc A X29 * X391 DVRY ←	DvryA X29 * X391 ↵	
View Apple Screen Mirroring Device Discovery Proxy Service	Esc A X29 DVRY ←	X391 ↵	
		DvryA X29 * X391 ↵	(Verbose mode 2/3)
mDNS Hostname			
Set mDNS hostname	Esc H X78 ZCON ←	ZconH• X78 ↵	
Clear mDNS hostname	Esc H•ZCON ←	ZconH• ↵	
View mDNS hostname	Esc HZCON ←	X78 ↵	
		ZconH• X78 ↵	(Verbose mode 2/3)

KEY:

X29 = LAN Port
- 1 = LAN Port A
- 2 = LAN Port B

X35 = Verbose response mode
- 0 = Clear (none) (default),
- 1 = Verbose mode,
- 2 = Tagged responses,
- 3 = Verbose + tagged responses

X36 = Layout mode
- 1 = Layout group #1 (default)
- 2 = Layout group #2
- 3 = Layout group #3
- 4 = Layout group #4
- 5 = Layout group #5

X37 = Current layout number (1-15)

X38 = HDMI input mode
- 0 = Disabled
- 1 = Enabled (default)

X39 = Apple screen mirroring mode
- 0 = Disabled
- 1 = Enabled (default)

X391 = Device discovery proxy service
- 0 = Disabled (default)
- 1 = Enabled

X78 = mDNS hostname (alpha-numeric up to 63 characters; No blanks, spaces, or special characters except hyphen (-)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Enable/disable HDMI input			
Enable HDMI input	`Esc` H1MODE←	ModeH`X40`↵	
Disable HDMI input	`Esc` H0MODE←	ModeH`X40`↵	
View HDMI input setting	`Esc` HMODE←	`X40`↵ ModeH`X40`↵	 (Verbose mode 2/3)
Show/hide HDMI client app controls			
Show HDMI client app controls	`Esc` V1APPC←	AppcV`X401`↵	
Hide HDMI client app controls	`Esc` V0APPC←	AppcV`X401`↵	
View HDMI client app controls setting	`Esc` VAPPC←	`X401`↵ AppcV`X401`↵	 (Verbose mode 2/3)
HDMI client app controls label			
Set client app HDMI controls label	`Esc` 1,`X41`NI←	Nmi1,`X41`↵	
Set label back to default (HDMI)	`Esc` 1,•NI←	Nmi1,HDMI↵	
View client app HDMI controls label	`Esc` 1NI←	`X41`↵ Nmi1,`X41`↵	 (Verbose mode 2/3)

KEY:

`X40` = HDMI input setting	0 = Disabled 1 = Enabled (default)
`X41` = Client app controls label	UTF-8 characters permitted, maximum of 16 characters
`X401` = HDMI client app controls	0 = Hide 1 = Show (default)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Access modes			
Set access mode	Esc M X42 SHAR ←	SharM X42 * X43 ↵	
View access mode	Esc MSHAR ←	X42 * X43 ↵	
		SharM X42 * X43 ↵	(Verbose mode 2/3)
Set confirmation-code mode	Esc M X44 PINC ←	PincM X44 ↵	
View confirmation-code mode	Esc MPINC ←	X44 ↵	
		PincM X44 ↵	(Verbose mode 2/3)
Set fixed login key code	Esc V X45 PINC ←	PincV• X45 ↵	
View current login key code	Esc VPINC ←	X45 ↵	
		PincV• X45 ↵	(Verbose mode 2/3)
Set random code interval	Esc I X46 PINC ←	PincI X46 ↵	
View random code interval	Esc IPINC ←	X46 ↵	
		PincI X46 ↵	(Verbose mode 2/3)
Set WebView password option	Esc M X47 WBSH ←	WbshM X47 ↵	
View WebView password option	Esc MWBSH ←	X47 ↵	
		WbshM X47 ↵	(Verbose mode 2/3)
Set WebView password	Esc P<password>WBSH ←	WbshP ↵	
Set WebView password back to default	Esc P•WBSH ←	WbshP**** ↵	Default = WebView
View WebView password	Esc PWBSH ←	**** ↵	If password is not set to default
		↵	If password is set to default
		WbshP**** ↵	(Verbose mode 2/3)
Set password	Esc PSWD*<username> ← <current password> ← <new password> ← <new password> ←	Pswd*<username> ↵	

NOTES:

- The PSWD command must use the carriage return character (hex 0D) and not the pipe character (|) as delimiters. This allows the pipe character to be part of a password, if desired.
- Password can be up to 128 characters. All human-readable characters are permitted. The password cannot be a single space. Passwords are case-sensitive.
- The PSWD command will return E35 if the account *<username>* does not exist.

KEY:

X42 = Access mode	1 = Full control 2 = Moderated 3 = Run-time (default)
X43 = Run time access mode	1 = Full control 2 = Moderated
X44 = Confirmation code mode	0 = None 1 = Fixed 2 = Random (default)
X45 = Login key code	Four digit value (default = 0000)
X46 = Random code interval	Four digit value (1 to 9998) in minutes Default = 9999 = random code generated after every collaboration session (after the last user disconnects)
X47 = WebView password option	0 = No access (disabled) 1 = No prompt for password (default) 2 = Custom password 3 = Follow confirmation code

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Standby Screen/Status bar configuration			
Set status bar mode	Esc MX48OSDL←	OsdlMX48↵	
View status bar mode	Esc MOSDL←	X48↵	
		OsdlMX48↵	(Verbose mode 2/3)
Set enable/disable status bar pop-up	Esc PX79OSDL←	OsdlPX79↵	
View status bar pop-up	Esc POSDL←	X79↵	
		OsdlPX79↵	(Verbose mode 2/3)
Set status bar pop-up timer	Esc TX791OSDL←	OsdlTX791↵	
View status bar pop-up timer	Esc TOSDL←	X791↵	
		OsdlTX791↵	(Verbose mode 2/3)
Set label	Esc LX49*X50OSDL←	OsdlLX49*X50↵	
View label	Esc LX49OSDL←	X50↵	
		OsdlLX49*X50↵	(Verbose mode 2/3)
Show label and value	Esc VX49*1OSDL←	OsdlVX49*X51↵	
Hide label and value	Esc VX49*0OSDL←	OsdlVX49*X51↵	
Query label visibility	Esc VX49OSDL←	X51↵	
		OsdlVX49*X51↵	(Verbose mode 2/3)
Set background color	Esc KX77OSDL←	OsdlKX77↵	
View background color	Esc KOSDL←	X117↵	
		OsdlKX77↵	(Verbose mode 2/3)
Set font color	Esc CX77OSDL←	OsdlCX77↵	
View font color	Esc COSDL←	X77↵	
		OsdlCX77↵	(Verbose mode 2/3)
Set still image transition effect	Esc EX53SSHW←	SshwEX53↵	
View still image transition effect	Esc ESSHW←	X53↵	
		SshwEX53↵	(Verbose mode 2/3)

KEY:

X48 = Status bar mode	0 = Never shown 1 = Always shown (default) 2 = Not shown only when one content is shared 3 = Not shown when one or more content is shared
X49 = Label	1 = Hostname 2 = IP address A 3 = IP address B, 4 = Login code 5 = Custom message 6 = Dynamic notifications/messages 7 = Clock
X50 = Label value (max 128 characters)	
X51 = Label visibility setting	0 = Hide 1 = Show (default)
X53 = Still image transition effect	0 = Cut (default) 1 = Fade
X77 = Color described as RGB Hex triplets (RRGGBB)	
X79 = Status bar pop-up	0 = Disabled 1 = Enabled (default)
X791 = Status bar pop-up timer	1s-300s in 1s increments (default = 20s)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Set still image transition time	Esc T X54 SSHW←	SshwT X54 ↵	
View still image transition time	Esc TSSHW←	X54 ↵	
		SshwT X54 ↵	(Verbose mode 2/3)
Set duration between still image sequences	Esc D X55 SSHW←	SshwD X55 ↵	
View duration between still image sequences	Esc DSSHW←	X55 ↵	
		SshwD X55 ↵	(Verbose mode 2/3)
Set file path for Standby/ Connected Image Gallery	Esc P X56 , <filepath>SSHW←	SshwP X56 , <filepath>↵	
Set file path back to default	Esc P X56 ,•SSHW←	SshwP X56 ,•↵	
View file path for Standby/ Connected Image Gallery	Esc P X56 SSHW←	<filepath>↵	
		SshwP X56 , <filepath>↵	(Verbose mode 2/3)
RS-232 Endpoint Control over LAN Port			
Set RS-232 port mode	Esc 01* X57 LRPT←	Lrpt01* X57 ↵	Set RS-232 port mode X57
View RS-232 port mode	Esc 01LRPT←	X57 ↵	View the current RS-232 port mode
		Lrpt01* X57 ↵	(Verbose mode 2/3)
Set serial port parameters	Esc 1* X58 , X59 , X60 , X61 CP←	Cpn01•Ccp X58 , X59 , X60 , X61 ↵	
View serial port parameters	Esc 1CP←	X58 , X59 , X60 , X61 ↵	
		Cpn01•Ccp X58 , X59 , X60 , X61 ↵	(Verbose mode 2/3)
Set UART start point	Esc X63 MD←	Pmd X63 ↵	Set port number start point
View UART start point	Esc MD←	X63 ↵	View port number start point
		Pmd X63 ↵	(Verbose mode 2/3)

KEY:

X54 = Still image transition time	500 to 3000ms increments (default = 1000ms)
X55 = Time between still image sequences	1000 to 60000ms in 1ms increments (default = 10000ms)
X56 = Gallery setting	1 = Standby image gallery 2 = Connected image gallery
X57 = Port mode	0 = disabled 1 = Built-in control (default) 2 = RS-232 Over LAN Pass-Thru
X58 = Baud rate	1200, 1800, 2400, 3600, 4800, 7200, 9600, 14400, 19200, 28800, 38400, 57600, 115200 (9600 = default)
X59 = Parity	**O**dd, **E**ven, **N**one, **M**ark, **S**pace (only the first letter is required; **N**one = default)
X60 = Data bits	7 or 8 (8 = default)
X61 = Stop bits	1 or 2 (1 = default)
X63 = Start Point for UART port	(Default = 2001)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Connections			
Query list of connected user information (up to 64 entries)	Esc L0SHAR←	X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ ... X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵↵ (Verbose mode 2/3): SharLX64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ ... X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵↵ *Response example:* 1406210687*user*1588900067*example_img.jpg*0*1*0*5000*10000*7307*1125039↵ 1842397516*Samsung*1982175097*example_img.jpg*0*1*0*5000*10000*7307*1350277↵ 1995721384*Mike**0******6981*0↵↵	
Query specific connected user information	Esc LX64SHAR←	X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ ... X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵↵ (Verbose mode 2/3): SharLX64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵ ... X64*X65*X66*X67*X68*X69*X70*X71*X72*X73*X75↵↵	
Query total number of users connected	Esc KSHAR←	X74↵ SharKX74↵	View total number of users connected (max 64) (Verbose mode 2/3)

KEY:

X64 = Connection ID (10 digits)

X65 = User name (apha-numeric text)

X66 = Stream ID (10 digits) (if connection is not streaming, this will be empty)

X67 = Content type (example: HDMI, App Window, AirPlay, etc)

X68 = Connection approved (0 = pending, 1 = approved)

X69 = X: x position of the window as a percentage of the screen width times 100

X70 = Y: y position of the window as a percentage of the screen height times 100

X71 = W: width of the window as a percentage of the screen width times 100

X72 = H: height of the window as a percentage of the screen height times 100

X73 = Connection age (in seconds)

X74 = Total number of users connected (integer – maximum 64)

X75 = Video port ID (7 digits)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Force user to stop sharing	`Esc S X64 SHAR ←`	`SharS X64 ↵`	Force user X64 to stop sharing content. If X64 = `0`, stop sharing all.
Force disconnect user	`Esc D X64 SHAR ←`	`SharD X64 ↵`	Force disconnection of user X64. If X64 = `0`, disconnect all
Approve user connection request (moderator mode only)	`Esc C X64 *1SHAR ←`	`SharC X64 *1↵`	Approve user request to connect
Reject user connection request (moderator mode only)	`Esc C X64 *0SHAR ←`	`SharC X64 *0↵`	Reject user request to connect
Approve user share request (moderator mode only)	`Esc R X64 *1SHAR ←`	`SharR X64 *1↵`	Approve user request to share content
Reject user share request (moderator mode only)	`Esc R X64 *0SHAR ←`	`SharR X64 *0↵`	Reject user request to share content
SIS-over-SSH			
NOTES: • The echo setting must match the type of SSH client used. With echo **enabled** (default), characters that are typed into the client window (such as PuTTY) are echoed in the window along with the SIS response, and are also sent to the server. • With echo **disabled**, the characters are not echoed but are simply sent to the server. • The echo setting applies only to current (not global) connection. • **Control systems should turn echo off after connecting.**			
Enable echo (default)	`Esc 1ECHO ←`	`Echo1↵`	Echo on: All data sent is echoed back to the sender, followed by the response. Extra carriage returns may also be received when echo is on.
Disable echo	`Esc 0ECHO ←`	`Echo0↵`	Echo off: only the response is sent to the sender.
View echo status	`Esc ECHO ←`	`X76 ↵`	View the echo setting
Set current port timeout	`Esc 0* X62 TC ←`	`Pti0* X62 ↵`	
View current port timeout	`Esc 0TC ←`	`X62 ↵`	
		`Pti0* X62 ↵`	(Verbose mode 2/3)
Set global port timeout	`Esc 1* X62 TC ←`	`Pti1* X62 ↵`	
View global port timeout	`Esc 1TC ←`	`X62 ↵`	

KEY:

X62 = Port timeout	`1` (= 10 seconds) to `65000` (650,000 seconds) (Default is `30` = 300 seconds = 5 minutes)
X64 = Connection ID (10 digits)	
X76 = Echo status	`0` = disabled `1` = enabled (default)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Reboot			
Reboot device	Esc 1BOOT←	Boot1↵	
Reset			
Reset to factory defaults	Esc ZXXX←	Zpx↵	Reset all device settings to default values, but retain user-loaded files and IP settings.
NOTE: This mode does not affect passwords and IP related settings such as IP address, subnet mask, gateway address, DHCP setting, and port mapping.			
Erase all files from flash memory	Esc ZFFF←	Zpf↵	Erase all files from flash (user) memory.
NOTE: This reset only removes files created in the user space, which includes those created by the backup and restore functions, software configuration tools, image captures, user logo files, and other user-supplied HTML files. Space used by firmware for internal operations such as saving non-volatile settings is not removed.			
Absolute reset, retaining IP settings	Esc ZY←	Zpy↵	Reset all device settings to default values and delete user-loaded files, but retain IP settings.
NOTE: • This reset removes all device settings, including the file system and passwords, but retains IP settings such as IP address, subnet mask, gateway address, DHCP setting, and port mapping (Telnet, Web, and direct access) in order to preserve communication with the device. This reset is recommended after a firmware update. • Admin password resets to `extron`			
IP system reset	Esc 1ZQQQ←	Zpq1↵	Reset IP settings only.
Absolute system reset	Esc ZQQQ←	Zpq↵	Reset all device and IP settings to default values and delete user-loaded files.
NOTE: Admin password resets to `extron`			

Unsolicited Responses

Response (Device to Host)		Description
Input selection and status	Pips1↵	HDMI window input is selected
	Pips0↵	HDMI window input is de-selected
	In1↵	HDMI pass-through input is selected
	In0↵	HDMI pass-through input is de-selected
	In00•X2↵	HDMI input signal presence changed
	Reconfig↵	Change of the input frequency
Detected format changed	Vtyp1*X4↵	Detected video format changed.
HDCP status	HdcpIX14↵	Change in the HDCP status of the HDMI input.
	HdcpOX14↵	Change in the HDCP status of the HDMI output.
Hot plug	HplgO↵	Hot plug occurs on the local HDMI output.
Contact closure status	CntcX25*X27↵	Change in the state of a contact closure input (X25)
Total number of users connected	SharKX74↵	Unsolicited response sent upon detection of the following:
		A new user has connected to the ShareLink Pro.
		A connected user has disconnected from the ShareLink Pro.
User changes	UserChg↵	Unsolicited response sent upon detection of the following:
		A connected user has started sharing content.
		A connected user has stopped sharing content.
		A user is requesting to connect to the ShareLink Pro (in moderator mode only).
		A connected user is requesting to share content (in moderator mode only).

KEY:

X2 = Input signal	0 = No input signal detected	1 = Input signal detected
X4 = Video format	0 = No signal 1 = DVI 2 = HDMI	
X14 = HDCP status	0 = Not connected 1 = Not HDCP Encrypted 2 = HDCP Encrypted	
X25 = Contact closure input	1 = Contact closure input 1 3 = Contact closure input 3	2 = Contact closure input 2 4 = Contact closure input 4
X27 = Contact/tally state	0 = Off (open) 1 = On (closed)	
X74 = Total number of users connected (integer – maximum 64)		

Expo and Streaming

NOTE: The following SIS commands for the Expo streaming feature require a LinkLicense.

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Expo Feature			
Enable/disable Expo feature	Esc P X104 SHAR ←	SharP X104 ↵	
View Expo feature	Esc PSHAR ←	X104 ↵	
		SharP X104 ↵	(Verbose mode 2/3)
View player state	Esc QSHAR ←	X105 ↵	
		SharQ X105 ↵	(Verbose mode 2/3)
Streaming Input			
View current stream statistics	Esc JBITR ←	<report in JSON format> ↵ ***Example:*** {“audio_bitrate_kbps”:0,” video_bitrate_kbps”:0, “peak_video_bitrate_kbps”:0, “stream_bitrate_kbps”:0, “received_packets”:0, “misordered_packets”:0, “duplicated_packets:0, “lost_packets”:0, “dropped_packets”:0, “jitter_ms”:0, “jitter_audio_ms”:0, “jitter_video_ms”:0} ↵	
View current stream mode	Esc SMOD ←	X80 ↵	
		Smod X80 ↵	(Verbose mode 2/3)
View current source audio sample rate	Esc AUSR ←	X81 ↵	
		Ausr X81 ↵	(Verbose mode 2/3)

KEY:

X104 = Expo feature	0 = Disabled (default) 1 = Continuous 2 = Temporary
X105 = Player state	1 = Standby 2 = Connected 3 = Expo 4 = Expo Standby 5 = Sharing
X80 = Stream mode	1 = Audio 2 = Video 3 = Audio and video (default)
X81 = Source audio sample rate	0 = Reserved 1 = Reserved 2 = 44.1 kHz 3 = 48 kHz

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Streaming Input Playback			
Start playback	Esc S X82 * X83 PLYR ←	PlyrS X82 * X83 ↵	Play channel X82 at speed X83.
Pause playback	Esc E X82 PLYR ←	PlyrE X82 * X83 ↵	
Stop playback	Esc O X82 PLYR ←	PlyrO X82 ↵	
View playback state	Esc Y X82 PLYR ←	X84 ↵	
		PlyrY X82 * X84 ↵	(Verbose mode 2/3)
Set loop play on	Esc R X82 *1PLYR ←	PlyrR1*1↵	
Set loop play off	Esc R X82 *0PLYR ←	PlyrR1*0↵	
View loop play status	Esc R X82 PLYR ←	X85 ↵	
		PlyrR1* X85 ↵	(Verbose mode 2/3)
Enable subtitles	Esc E X82 *1SUBT ←	SubtE1*1↵	
Disable subtitles	Esc E X82 *0SUBT ←	SubtE1*0↵	
View subtitle status	Esc E X82 SUBT ←	X85 ↵	
		SubtE1* X85 ↵	(Verbose mode 2/3)
Select next channel	Esc N X82 PLYR ←	PlyrN1* X86 ↵ PlyrT1* X86 ↵ PlyrS1* X83 ↵	Response returns the current channel number (T) and the current playback speed (S).
Select previous channel	Esc P X82 PLYR ←	PlyrP1* X86 ↵ PlyrT1* X86 ↵ PlyrS1* X83 ↵	Response returns the current channel number (T) and the current playback speed (S).
View current timecode	Esc K X82 PLYR ←	X87 ↵	
		PlyrK X82 * X87 ↵	(Verbose mode 2/3)
View current clip length	Esc Z X82 PLYR ←	X87 ↵	X87 = Timecode in HH:MM:SS:DD format
		PlyrZ X87 ↵	(Verbose mode 2/3)
Seek by offset	Esc J X82 * X88 PLYR ←	PlyrJ X82 * X88 ↵	
Seek to timecode	Esc K X82 * X87 PLYR ←	PlyrK X82 * X87 ↵	Set playback position
View current timecode	Esc K X82 PLYR ←	X87 ↵	
		PlyrK X82 * X87 ↵	(Verbose mode 2/3)
Load media item path	Esc U X82 * X89 PLYR ←	PlyrU X82 * X89 ↵	
View current media path	Esc U X82 PLYR ←	X89 ↵	
		PlyrU X82 * X89 ↵	(Verbose mode 2/3)

KEY:

X82 = Playback channel (always 1 for single channel devices)

X83 = Playback speed — 1 = normal speed

X84 = Player state — 0 = Stop, 1 = Play, 2 = Pause

X85 = Loop play status — 0 = Off/disable, 1 = On/enable

X86 = Track number (1 to maximum tracks currently defined)

X87 = Timecode (HH:MM:SS:DD, where DD is decimal seconds up to nine digits)

X88 = Time (in seconds)

X89 = File path

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Streaming Input Channel (Preset)			
Short Form			
View current preset	T	X93↵	Get current channel number. Response is 00 if no valid channel preset is loaded.
Recall preset by reference	X93T	TvprTX82*X93↵	Load channel X93
Next preset	+T	TvprTX82*X93↵	Channel up
Previous preset	-T	TvprTX82*X93↵	Channel down
Long Form			
View current preset	Esc TX82TVPR←	X93↵	Get current channel number. Response is 00 if no valid channel preset is loaded.
Recall preset by reference	Esc TX82*X93TVPR←	TvprTX82*X93↵	Load channel X93
Next preset	Esc TX82+TVPR←	TvprTX82*X93↵	Channel up
Previous preset	Esc TX82-TVPR←	TvprTX82*X93↵	Channel down
View all presets	Esc GTVPR←	[{…},{…},…]↵ *Example*: [{"preset":1, "uri":file:///clips/clip1.mp4,"name":"Chan1"},{…},…]	Get channel list
Save URI to preset	Esc UX93*X90TVPR←	TvprUX93*X91↵	Save URI to channel X93
Save current URI to preset	Esc SX82*X93TVPR←	TvprSX82*X93↵	Save current URI to channel X93
Import channel preset	Esc I*X89TVPR←	TvprI*X89↵	
Export channel preset	Esc E*X89TVPR←	TvprE*X89↵	
Delete preset	Esc XX93TVPR←	TvprXX93↵	
Delete all presets	Esc X*0TVPR←	TvprX↵	

KEY:

X82 = Playback channel (always 1 for single channel devices)

X89 = File path (file:///folder/filename, network path, network port path or RTSP stream URI)
X90 = Media URI (Uniform Resource Identifier). The prefix "file:///" is optional.
X91 = Preset number, and Media URI — Example: [{"preset":1, "uri":file:///clips/clip1.mp4, "name":"Chan1"}, …]

X93 = Channel number — 1 - 999

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
Network Shares			

Add a network share	`Esc A{"service":"X106",`
	`"mountpoint":"X107",`
	`"username":"X108",`
	`"password":"X109",`
	`"options":"X110",`
	`"reconnect":"X111",`
	`"extron_share":"X112"}`
	`XNAS←`
		`XnasA<parameters & status in JSON format>↵`	Add or mount a network share
			
			Example:
			`XnasA{"username":"user1", "password":"****", "options":"ro", …}↵`

Unmount (remove) a share	`Esc X*X107XNAS←`	`XnasX*X107↵`	
Unmount (remove) all shares	`Esc X*0XNAS←`	`XnasX*0↵`	

View all mounted shares	`Esc GXNAS←`	`[list of shares in JSON format]↵`
		
		Example:
		`[{"options": "", "username": "", "password": "", "type": "nfs", "service": "10.0.0.0://home/bob/Videos", "mountpoint": "10.0.0.0", "extron_share": false, "status": "unavailable", "recon-nect": true}, {…}] ↵`
		
		(Verbose mode 2/3):
		`XnasG[list of shares in JSON format]↵`	

KEY:

X106 = Service (network path of server) **CIFS**: "`\\\\myServer\\myShare\\myDir`"
NFS: "`myServer:/myDir`"

X107 = Mountpoint
X108 = Username
X109 = Password
X110 = Options
X111 = Boolean for `reconnect`
X112 = Boolean for `extron_share`

NOTE: The boolean value should be "true" or "false." It does not require quotes and must be all lowercase.

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
View File-System Directory Structure			
View file-system directory structure	[Esc][X113]*[X114]TREE←		
	[Esc]TREE←	/a↵ /a/b↵ /a/b/c↵ /Connected↵ /Connected/ Thumbnails↵ /media↵ /nortxe-backup↵ /shares↵ /Standby↵ /Standby/ Thumbnails↵ /tmp↵↵	
	[Esc]0TREE←	/a↵ /Connected↵ /media↵ /nortxe-backup↵ /shares↵ /Standby↵ /tmp↵	
	[Esc]1*/aTREE←	/b↵ /b/c↵↵ Tree1*/a*/b↵ /b/c/↵↵	(Verbose mode 2/3)

KEY:

[X113] = Recursion (optional)
- 0 = Disabled
- 1 = Enabled (default)

[X114] = Directory path (optional)

Command	ASCII Command (Host to Device)	Response (Device to Host)	Additional Description
SAP (Session Announcement Protocol)			
View list of discovered streams	Esc XSAP←	[list of discovered streams in JSON format]↵	*Example*: [{“sdp”:“”, “source”:“”, …}, {“sdp”:“”, “source”:“”, …}]↵
		Xsap[{}, {}, …]↵	(Verbose mode 2/3)
Network Status and Statistics			
View network status	Esc X115NWKS←	<report in JSON format>↵	*Example*: {“collisions”: 0, “rx_bytes”: 104222342, “rx_dropped”: 2197, “rx_errors”: 0, “rx_packets”: 98850, “tx_bytes”: 20497039, “tx_dropped”: 0, “tx_errors”: 0, “tx_packets”: 42050}↵
		NwksX115*<report in JSON format>↵	(Verbose mode 2/3)
Buffering			
Set player buffering	Esc EX82*X85PBUF←	PbufEX82*X85↵	
View player buffering status	Esc EX82PBUF←	X85↵	
		PbufEX82*X85↵	(Verbose mode 2/3)
Set initial play threshold	Esc IX82*X116PBUF←	PbufIX82*X116↵	
View initial play threshold	Esc IX82PBUF←	X116↵	
		PbufIX82*X116↵	(Verbose mode 2/3)
Set rebuffering play threshold	Esc RX82*X117PBUF←	PbufRX82*X117↵	
View rebuffering play threshold	Esc RX82PBUF←	X117↵	
		PbufRX82*X117↵	(Verbose mode 2/3)
Prefer RTSP Multicast			
Prefer RTSP Multicast	Esc QX82*X85PLYR←	PlyrQX82*X85↵	Request multicast stream first and revert to requesting a unicast stream if necessary.
View RTSP Multicast preference	Esc QX82PLYR←	X85↵	
		PlyrQX82*X85↵	(Verbose mode 2/3)

KEY:

X82 = Player channel (Always 1 for single channel devices)

X85 = Buffering and RTSP Multicast setting 0 = Off/Disabled 1 = On/Enabled

X115 = Network interface 1 = LAN A 2 = LAN B

X116 = Initial play threshold (sec)

X117 = Rebuffering play threshold (sec)

Reference Information

This section covers the following:

- **Mounting**
- **Firmware Download**
- **Firewall Traversal**
- **Troubleshooting**

Mounting

The ShareLink Pro 1000 can be placed on a tabletop, or mounted in a rack or underneath a desk.

> **NOTE:** Do not block or restrict air ventilation holes on the top and sides of the ShareLink Pro 1000.

Tabletop Mounting

Attach the provided rubber feet to the bottom four corners of the enclosure.

Furniture Mounting

Go to **www.extron.com** for a list of available furniture mounting kits. To install this device to furniture, follow the mounting kit instructions.

Rack Mounting

UL Guidelines for Rack Mounting

The following Underwriters Laboratories (UL) guidelines are relevant to the safe installation of this product in a rack:

- **Elevated operating ambient temperature** — If the unit is installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient temperature. Therefore, install the equipment in an environment compatible with the maximum ambient temperature (TMA = +122 °F, +40 °C) specified by Extron.
- **Reduced air flow** — Install the equipment in the rack so that safe operation and adequate air flow is provided to the unit.
- **Mechanical loading** — Mount the equipment in the rack so that a hazardous condition is not achieved due to uneven mechanical loading.
- **Circuit overloading** — Connect the equipment to the supply circuit and consider the effect that circuit overloading might have on overcurrent protection and supply wiring. Consider the equipment nameplate ratings when addressing this concern.

- **Reliable earthing (grounding)** — Maintain reliable grounding of rack-mounted equipment. Pay particular attention to supply connections other than direct connections.

Rack Mounting Procedure

The ShareLink Pro 1000 can be mounted in a rack, using an optional Extron rack mounting kit (see **www.extron.com** for part numbers). Follow the instructions provided with the kit.

Firmware Download

To download the latest firmware for the ShareLink Pro 1000:

1. On the Extron website, **www.extron.com**, go to the **Download** tab and click **Firmware** (see figure 19, ❶).

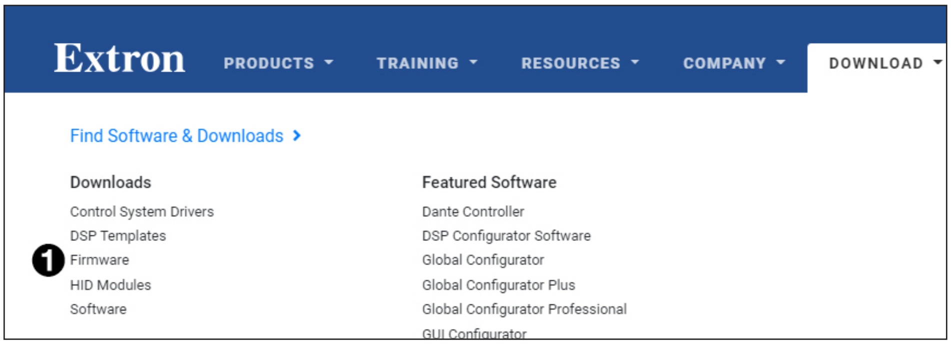


Figure 19. Firmware Link on the Download Tab

2. In the Download Center screen, navigate to the ShareLink Pro 1000 (see figure 20, ❶).

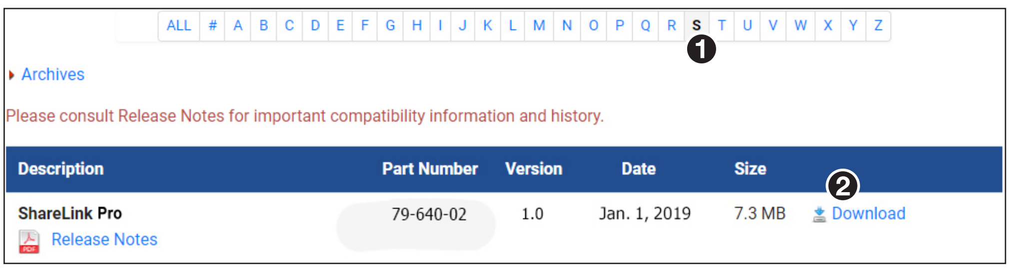


Figure 20. Firmware Download Center

3. Ensure the available firmware version is a later version than the current one on the device, and click the **Download** link (❷).

> **NOTE:** The firmware release notes provide details about the changes between different firmware versions. The file can be downloaded from the same page as the firmware.

4. Submit any required information to start the download. Note where the file is saved.

> **NOTE:** For instructions on loading new firmware on the device, see **Firmware** on page 16.

Firewall Traversal

In order for the ShareLink Pro 1000 gateway to function correctly with the ShareLink Pro applications, it is recommended that the following firewall rules be used to allow traffic to pass. The table below details the ports and their functions.

Port	Type	Direction	Function	Open	Notes
53	UDP/ TCP	Both	DNS	Required	This port is used for communication with a DNS server.
67	UDP	Both	DHCP	Required	This port is used for communication with a DHCP server.
68	UDP	Both	DHCP	Required	This port is used for communication with a DHCP server.
80	TCP	Both	HTTP	Required	This port is used to access the default webpage.
123	UDP	Both	NTP	Required	This port is used for communication with an NTP server.
161	UDP	Both	SNMP	Optional	This port is used for SNMP management.
443	TCP	Both	HTTPS	Required	This port is used for accessing the admin configuration webpage.
1041	UDP	Both	GVE Discovery	Optional	This port is used for discovery of ShareLink Pro devices by GVE software.
1230	UDP	Outbound	Discovery	Required	This port is used for discovery of ShareLink Pro devices by software client applications.
1231	UDP	Inbound	Discovery	Required	This port is used for discovery of ShareLink Pro devices by software client applications.
4502	TCP	Both	GVE	Optional	This port is used for GVE management.
4503	TCP	Both	GVE	Optional	This port is used for GVE management.
4522	TCP	Both	PCS file transfer	Required	This port is used for configuration.
4523	TCP	Both	PCS config.	Required	This port is used for configuration.
4534	TCP	Both	ID messaging service	Optional	This port is used for Extron Control System messaging.
5353	UDP	Both	Apple Screen Mirroring Discovery	Required	This port is used by Bonjour/Avahi network discovery for Apple Screen Mirroring discovery.
7000	TCP	Both	Apple Screen Mirroring	Required	This port is used for Apple Screen Mirroring.
7100	TCP	Both	Apple Screen Mirroring	Required	This port is used for Apple Screen Mirroring.
NOTE: mDNS/Multicast must be enabled on the network in order for Apple Screen Mirroring to function properly					
1000-65384	UDP	Outbound	Video streaming	Required	These ports are used for streaming video data to the ShareLink Pro. Port is negotiated at run-time in the range 1000 to 65384 (randomly).

Port	Type	Direction	Function	Open	Notes
1000-65384	UDP	Outbound	Audio streaming	Required	These ports are used for streaming audio data to the ShareLink Pro. Port is negotiated at run-time in the range 1000 to 65384 (randomly).
22023	TCP	Inbound	Control	Required	This port is used for controlling the ShareLink Pro via SIS-over-SSH.

Troubleshooting

The following table gives recommended actions to solve problems that may occur during setup or operation. If the problem persists after performing the recommended action, contact Extron Technical Support.

Problem	Actions
(Windows only) Audio level lowers while using the Mirror Screen or Mirror Application features	Open the **Sound** settings on your PC, and check the following: • Check that Dolby Digital Audio is disabled in your PC's Sounds settings as follows: **1.** Open the **Sounds** settings and go to the **Playback** tab. **2.** Right-click **Speakers/Headphones** and click **Properties**. **3.** Go to the **Enhancements** tab. **4.** Make sure Dolby Digital Audio is disabled. • In the **Communications** tab, make sure the setting for “**When Windows detects communication activity**” is set to **Do nothing**.
(Windows only) No audio for the Mirror Screen or Mirror Application features	• Make sure that the ShareLink Pro application is not muted in the PC’s volume mixer. • In the **Recording** tab (within the PC’s **Sounds** settings), make sure that **Stereo Mix** is enabled and set as default device. • In the **Playback** tab, check that **Speaker** is enabled and set as default device. • In the **Recording** and **Playback** tabs, check that the ShareLink Pro Audio Driver is enabled. • Download and install the ShareLink Pro 1000 software from the Extron website (see **page 10** for instructions).
(Windows only) Extron Audio Driver has no audio support after being installed	• In the **Recording** and **Playback** tabs, check that the ShareLink Pro Audio Driver is enabled and set as default device when the ShareLink Pro application is running. • Download and install the ShareLink Pro 1000 software from the Extron website.
(iOS only) The ShareLink Pro device is not listed in the Screen Mirroring list of devices.	• Make sure the iOS device is on the same network as the ShareLink Pro device. • In PCS, go to **General Settings**, then the **Access/Role Setup** tab and make sure **IOS Airplay** is enabled. • Make sure the specific network ports required for Apple Screen Mirroring are open (see **Firewall Traversal** on the previous page for port requirements).

Problem	Actions
4K/60 is not outputting on 4K/60 capable display	• Test the connection using a shorter HDMI cable. • In PCS, go to **`EDID Minder`**. In the `EDID Assignment` panel, make sure `HDMI Pass-Through` is set to 4K @60Hz. • In PCS, go to **`Input/Output Config`**. In the `Output Configuration` panel, set the `Format` to the appropriate color space for the display.
(Windows only) Mirror Screen and Mirror Application features are slow or not working	• Make sure that the graphics drivers are up to date. • Download and reinstall graphics drivers directly from the driver website (such as Intel®, NVIDIA® GeForce®, AMD Radeon™, etc).
(iOS only) The Share Website feature on the iOS app shows a blank page	Make sure the device is connected to the internet.
Windows 8.1 unable to install .net 4.6.2	Install Windows update KB2919355.
Contact/Tally ports are not responsive or are slow to respond	Check termination of Contact/Tally ports.

Extron Warranty

Extron warrants this product against defects in materials and workmanship for a period of three years from the date of purchase. In the event of malfunction during the warranty period attributable directly to faulty workmanship and/or materials, Extron will, at its option, repair or replace said products or components, to whatever extent it shall deem necessary to restore said product to proper operating condition, provided that it is returned within the warranty period, with proof of purchase and description of malfunction to:

USA, Canada, South America, and Central America: Extron 1230 South Lewis Street Anaheim, CA 92805 U.S.A.	**Asia:** Extron Asia Pte Ltd 135 Joo Seng Road, #04-01 PM Industrial Bldg. Singapore 368363 Singapore	**Japan:** Extron Japan Kyodo Building, 16 Ichibancho Chiyoda-ku, Tokyo 102-0082 Japan
Europe: Extron Europe Hanzeboulevard 10 3825 PH Amersfoort The Netherlands	**China:** Extron China 686 Ronghua Road Songjiang District Shanghai 201611 China	**Middle East:** Extron Middle East Dubai Airport Free Zone F13, PO Box 293666 United Arab Emirates, Dubai
Africa: Extron South Africa 3rd Floor, South Tower 160 Jan Smuts Avenue Rosebank 2196, South Africa		

This Limited Warranty does not apply if the fault has been caused by misuse, improper handling care, electrical or mechanical abuse, abnormal operating conditions, or if modifications were made to the product that were not authorized by Extron.

NOTE: If a product is defective, please call Extron and ask for an Application Engineer to receive an RA (Return Authorization) number. This will begin the repair process.

USA:	714.491.1500 or 800.633.9876	**Asia**:	65.6383.4400
Europe:	31.33.453.4040 or 800.3987.6673	**Japan**:	81.3.3511.7655
Africa:	27.11.447.6162	**Middle East**:	971.4.299.1800

Units must be returned insured, with shipping charges prepaid. If not insured, you assume the risk of loss or damage during shipment. Returned units must include the serial number and a description of the problem, as well as the name of the person to contact in case there are any questions.

Extron makes no further warranties either expressed or implied with respect to the product and its quality, performance, merchantability, or fitness for any particular use. In no event will Extron be liable for direct, indirect, or consequential damages resulting from any defect in this product even if Extron has been advised of such damage.

Please note that laws vary from state to state and country to country, and that some provisions of this warranty may not apply to you.